

The *RLC* Circuit

INTRODUCTION

In Chap. 8 we studied circuits which contained only **one** energy storage element, combined with a passive network which partly determined how long it took either the capacitor or the inductor to charge/discharge. The differential equations which resulted from analysis were always first-order. In this chapter, we consider more complex circuits which contain **both** an inductor **and** a capacitor. The result is a **second-order** differential equation for any voltage or current of interest. What we learned in Chap. 8 is easily extended to the study of these so-called *RLC* circuits, although now we need **two** initial conditions to solve each differential equation. Such circuits occur routinely in a wide variety of applications, including oscillators and frequency filters. They are also very useful in modeling a number of practical situations, such as automobile suspension systems, temperature controllers, and even the response of an airplane to changes in elevator and aileron positions.

9.1 THE SOURCE-FREE PARALLEL CIRCUIT

There are two basic types of *RLC* circuits: *parallel connected*, and *series connected*. We could start with either, but somewhat arbitrarily choose to begin by considering parallel *RLC* circuits. This particular combination of ideal elements is a reasonable model for portions of many communication networks. It represents, for example, an important part of certain electronic amplifiers found in radios, and enables the amplifiers to produce a large voltage amplification over a narrow band of signal frequencies (with almost zero amplification outside this band).

Just as we did with *RL* and *RC* circuits, we first consider the natural response of a parallel *RLC* circuit, where one or both of the

KEY CONCEPTS

Resonant Frequency and Damping Factor of Series and Parallel *RLC* Circuits

Overdamped Response

Critically Damped Response

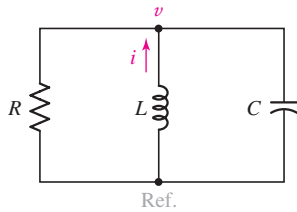
Underdamped Response

Making Use of Two Initial Conditions

Complete (Natural + Forced) Response of *RLC* Circuits

Representing Differential Equations Using Op Amp Circuits





■ FIGURE 9.1 The source-free parallel RLC circuit.

energy storage elements have some nonzero initial energy (the origin of which for now is unimportant). This is represented by the inductor current and the capacitor voltage, both specified at $t = 0^+$. Once we're comfortable with this part of RLC circuit analysis, we can easily include dc sources, switches, or step sources in the circuit. Then we find the total response, which will be the sum of the natural response and the forced response.

Frequency selectivity of this kind enables us to listen to the transmission of one station while rejecting the transmission of any other station. Other applications include the use of parallel RLC circuits in frequency multiplexing and harmonic-suppression filters. However, even a simple discussion of these principles requires an understanding of such terms as *resonance*, *frequency response*, and *impedance*, which we have not yet discussed. Let it suffice to say, therefore, that an understanding of the natural behavior of the parallel RLC circuit is fundamentally important to future studies of communications networks and filter design, as well as many other applications.

When a *physical* capacitor is connected in parallel with an inductor and the capacitor has associated with it a finite resistance, the resulting network can be shown to have an equivalent circuit model like that shown in Fig. 9.1. The presence of this resistance can be used to model energy loss in the capacitor; over time, all real capacitors will eventually discharge, even if disconnected from a circuit. Energy losses in the physical inductor can also be taken into account by adding an ideal resistor (in series with the ideal inductor). For simplicity, however, we restrict our discussion to the case of an essentially ideal inductor in parallel with a “leaky” capacitor.

Obtaining the Differential Equation for a Parallel RLC Circuit

In the following analysis we will assume that energy may be stored initially in both the inductor and the capacitor; in other words, nonzero initial values of both inductor current and capacitor voltage may be present. With reference to the circuit of Fig. 9.1, we may then write the single nodal equation

$$\frac{v}{R} + \frac{1}{L} \int_{t_0}^t v \, dt' - i(t_0) + C \frac{dv}{dt} = 0 \quad [1]$$

Note that the minus sign is a consequence of the assumed direction for i . We must solve Eq. [1] subject to the initial conditions

$$i(0^+) = I_0 \quad [2]$$

and

$$v(0^+) = V_0 \quad [3]$$

When both sides of Eq. [1] are differentiated once with respect to time, the result is the linear second-order homogeneous differential equation

$$C \frac{d^2v}{dt^2} + \frac{1}{R} \frac{dv}{dt} + \frac{1}{L}v = 0 \quad [4]$$

whose solution $v(t)$ is the desired natural response.

Solution of the Differential Equation

There are a number of interesting ways to solve Eq. [4]. Most of these methods we will leave to a course in differential equations, selecting only the quickest and simplest method to use now. We will assume a solution, relying upon our

intuition and modest experience to select one of the several possible forms that are suitable. Our experience with first-order equations might suggest that we at least try the exponential form once more. Thus, we *assume*

$$v = Ae^{st} \quad [5]$$

being as general as possible by allowing A and s to be complex numbers if necessary. Substituting Eq. [5] in Eq. [4], we obtain

$$CA s^2 e^{st} + \frac{1}{R} A s e^{st} + \frac{1}{L} A e^{st} = 0$$

or

$$A e^{st} \left(C s^2 + \frac{1}{R} s + \frac{1}{L} \right) = 0$$

In order for this equation to be satisfied for all time, at least one of the three factors must be zero. If either of the first two factors is set equal to zero, then $v(t) = 0$. This is a trivial solution of the differential equation which cannot satisfy our given initial conditions. We therefore equate the remaining factor to zero:

$$C s^2 + \frac{1}{R} s + \frac{1}{L} = 0 \quad [6]$$

This equation is usually called the *auxiliary equation* or the **characteristic equation**, as we discussed in Sec. 8.1. If it can be satisfied, then our assumed solution is correct. Since Eq. [6] is a quadratic equation, there are two solutions, identified as s_1 and s_2 :

$$s_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \quad [7]$$

and

$$s_2 = -\frac{1}{2RC} - \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}} \quad [8]$$

If *either* of these two values is used for s in the assumed solution, then that solution satisfies the given differential equation; it thus becomes a valid solution of the differential equation.

Let us assume that we replace s by s_1 in Eq. [5], obtaining

$$v_1 = A_1 e^{s_1 t}$$

and, similarly,

$$v_2 = A_2 e^{s_2 t}$$

The former satisfies the differential equation

$$C \frac{d^2 v_1}{dt^2} + \frac{1}{R} \frac{dv_1}{dt} + \frac{1}{L} v_1 = 0$$

and the latter satisfies

$$C \frac{d^2 v_2}{dt^2} + \frac{1}{R} \frac{dv_2}{dt} + \frac{1}{L} v_2 = 0$$

Adding these two differential equations and combining similar terms, we have

$$C \frac{d^2 (v_1 + v_2)}{dt^2} + \frac{1}{R} \frac{d(v_1 + v_2)}{dt} + \frac{1}{L} (v_1 + v_2) = 0$$

Linearity triumphs, and it is seen that the *sum* of the two solutions is also a solution. We thus have the general form of the natural response

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad [9]$$

where s_1 and s_2 are given by Eqs. [7] and [8]; A_1 and A_2 are two arbitrary constants which are to be selected to satisfy the two specified initial conditions.

Definition of Frequency Terms

The form of the natural response as given in Eq. [9] offers little insight into the nature of the curve we might obtain if $v(t)$ were plotted as a function of time. The relative amplitudes of A_1 and A_2 , for example, will certainly be important in determining the shape of the response curve. Furthermore, the constants s_1 and s_2 can be real numbers or conjugate complex numbers, depending upon the values of R , L , and C in the given network. These two cases will produce fundamentally different response forms. Therefore, it will be helpful to make some simplifying substitutions in Eq. [9].

Since the exponents $s_1 t$ and $s_2 t$ must be dimensionless, s_1 and s_2 must have the unit of some dimensionless quantity “per second.” From Eqs. [7] and [8] we therefore see that the units of $1/2RC$ and $1/\sqrt{LC}$ must also be s^{-1} (i.e., seconds $^{-1}$). Units of this type are called **frequencies**.

Let us define a new term, ω_0 (omega-sub-zero, or just omega-zero):

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad [10]$$

and reserve the term **resonant frequency** for it. On the other hand, we will call $1/2RC$ the **neper frequency**, or the **exponential damping coefficient**, and represent it by the symbol α (alpha):

$$\alpha = \frac{1}{2RC} \quad [11]$$

This latter descriptive expression is used because α is a measure of how rapidly the natural response decays or damps out to its steady, final value (usually zero). Finally, s , s_1 , and s_2 , which are quantities that will form the basis for some of our later work, are called **complex frequencies**.

We should note that s_1 , s_2 , α , and ω_0 are merely symbols used to simplify the discussion of *RLC* circuits; they are not mysterious new properties of any kind. It is easier, for example, to say “*alpha*” than it is to say “*the reciprocal of 2RC*.”

Let us collect these results. The natural response of the parallel *RLC* circuit is

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad [9]$$

where

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \quad [12]$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} \quad [13]$$

$$\alpha = \frac{1}{2RC} \quad [11]$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad [10]$$

and A_1 and A_2 must be found by applying the given initial conditions.

We note two basic scenarios possible with Eqs. [12] and [13] depending on the relative sizes of α and ω_0 (dictated by the values of R , L , and C). If $\alpha > \omega_0$, s_1 and s_2 will both be real numbers, leading to what is referred to as an **overdamped response**. In the opposite case, where $\alpha < \omega_0$, both s_1 and s_2 will have nonzero imaginary components, leading to what is known as an **underdamped response**. Both of these situations are considered separately in the following sections, along with the special case of $\alpha = \omega_0$, which leads to what is called a **critically damped response**. We should also note that the general response comprised by Eqs. [9] through [13] describes not only the voltage but all three branch currents in the parallel RLC circuit; the constants A_1 and A_2 will be different for each, of course.

The ratio of α to ω_0 is called the *damping ratio* by control system engineers and is designated by ζ (zeta).

Overdamped: $\alpha > \omega_0$
 Critically damped: $\alpha = \omega_0$
 Underdamped: $\alpha < \omega_0$



EXAMPLE 9.1

Consider a parallel RLC circuit having an inductance of 10 mH and a capacitance of 100 μ F. Determine the resistor values that would lead to overdamped and underdamped responses.

We first calculate the resonant frequency of the circuit:

$$\omega_0 = \sqrt{\frac{1}{LC}} = \sqrt{\frac{1}{(10 \times 10^{-3})(100 \times 10^{-6})}} = 10^3 \text{ rad/s}$$

An *overdamped* response will result if $\alpha > \omega_0$; an *underdamped* response will result if $\alpha < \omega_0$. Thus,

$$\frac{1}{2RC} > 10^3$$

and so

$$R < \frac{1}{(2000)(100 \times 10^{-6})}$$

or

$$R < 5 \Omega$$

leads to an overdamped response; $R > 5 \Omega$ leads to an underdamped response.

PRACTICE

9.1 A parallel RLC circuit contains a 100 Ω resistor and has the parameter values $\alpha = 1000 \text{ s}^{-1}$ and $\omega_0 = 800 \text{ rad/s}$. Find (a) C ; (b) L ; (c) s_1 ; (d) s_2 .

Ans: 5 μ F; 312.5 mH; -400 s^{-1} ; -1600 s^{-1} .

9.2 THE OVERDAMPED PARALLEL RLC CIRCUIT

A comparison of Eqs. [10] and [11] shows that α will be greater than ω_0 if $LC > 4R^2C^2$. In this case the radical used in calculating s_1 and s_2 will be real, and both s_1 and s_2 will be real. Moreover, the following inequalities

$$\sqrt{\alpha^2 - \omega_0^2} < \alpha$$

$$\left(-\alpha - \sqrt{\alpha^2 - \omega_0^2}\right) < \left(-\alpha + \sqrt{\alpha^2 - \omega_0^2}\right) < 0$$

may be applied to Eqs. [12] and [13] to show that both s_1 and s_2 are *negative* real numbers. Thus, the response $v(t)$ can be expressed as the (algebraic) sum of two decreasing exponential terms, both of which approach zero as time increases. In fact, since the absolute value of s_2 is larger than that of s_1 , the term containing s_2 has the more rapid rate of decrease, and, for large values of time, we may write the limiting expression

$$v(t) \rightarrow A_1 e^{s_1 t} \rightarrow 0 \quad \text{as } t \rightarrow \infty$$

The next step is to determine the arbitrary constants A_1 and A_2 in conformance with the initial conditions. We select a parallel RLC circuit with $R = 6 \, \Omega$, $L = 7 \, \text{H}$, and, for ease of computation, $C = \frac{1}{42} \, \text{F}$. The initial energy storage is specified by choosing an initial voltage across the circuit $v(0) = 0$ and an initial inductor current $i(0) = 10 \, \text{A}$, where v and i are defined in Fig. 9.2.

We may easily determine the values of the several parameters

$$\begin{aligned} \alpha &= 3.5 & \omega_0 &= \sqrt{6} \\ s_1 &= -1 & s_2 &= -6 \end{aligned} \quad (\text{all } s^{-1})$$

and immediately write the general form of the natural response

$$v(t) = A_1 e^{-t} + A_2 e^{-6t} \quad [14]$$

Finding Values for A_1 and A_2

Only the evaluation of the two constants A_1 and A_2 remains. If we knew the response $v(t)$ at two different values of time, these two values could be substituted in Eq. [14] and A_1 and A_2 easily found. However, we know only one instantaneous value of $v(t)$,

$$v(0) = 0$$

and, therefore,

$$0 = A_1 + A_2 \quad [15]$$

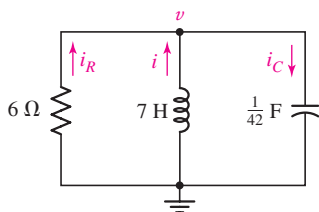


We can obtain a second equation relating A_1 and A_2 by taking the derivative of $v(t)$ with respect to time in Eq. [14], determining the initial value of this derivative through the use of the remaining initial condition $i(0) = 10$, and equating the results. So, taking the derivative of both sides of Eq. [14],

$$\frac{dv}{dt} = -A_1 e^{-t} - 6A_2 e^{-6t}$$

and evaluating the derivative at $t = 0$,

$$\left. \frac{dv}{dt} \right|_{t=0} = -A_1 - 6A_2$$



■ FIGURE 9.2 A parallel RLC circuit used as a numerical example. The circuit is overdamped.

we obtain a second equation. Although this may appear to be helpful, we do not have a numerical value for the initial value of the derivative, so we do not yet have two equations in two unknowns . . . Or do we? The expression dv/dt suggests a capacitor current, since

$$i_C = C \frac{dv}{dt}$$

Kirchhoff's current law must hold at any instant in time, as it is based on conservation of electrons. Thus, we may write

$$-i_C(0) + i(0) + i_R(0) = 0$$

Substituting our expression for capacitor current and dividing by C ,

$$\left. \frac{dv}{dt} \right|_{t=0} = \frac{i_C(0)}{C} = \frac{i(0) + i_R(0)}{C} = \frac{i(0)}{C} = 420 \text{ V/s}$$

since zero initial voltage across the resistor requires zero initial current through it. We thus have our second equation,

$$420 = -A_1 - 6A_2 \quad [16]$$

and simultaneous solution of Eqs. [15] and [16] provides the two amplitudes $A_1 = 84$ and $A_2 = -84$. Therefore, the final numerical solution for the natural response of this circuit is

$$v(t) = 84(e^{-t} - e^{-6t}) \quad \text{V} \quad [17]$$

For the remainder of our discussions concerning RLC circuits, we will always require two initial conditions in order to completely specify the response. One condition will usually be very easy to apply—either a voltage or current at $t = 0$. It is the second condition that usually requires a little effort. Although we will often have both an initial current and an initial voltage at our disposal, one of these will need to be applied indirectly through the derivative of our assumed solution.

EXAMPLE 9.2

Find an expression for $v_C(t)$ valid for $t > 0$ in the circuit of Fig. 9.3a.

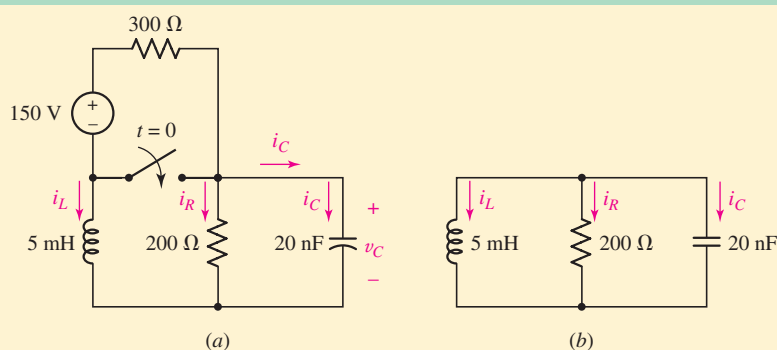


FIGURE 9.3 (a) An RLC circuit that becomes source-free at $t = 0$. (b) The circuit for $t > 0$, in which the 150 V source and the 300 Ω resistor have been shorted out by the switch, and so are of no further relevance to v_C .

► Identify the goal of the problem.

We are asked to find the capacitor voltage after the switch is thrown. This action leads to no sources remaining connected to either the inductor or the capacitor.

(Continued on next page)

► **Collect the known information.**

After the switch is thrown, the capacitor is left in parallel with a $200\ \Omega$ resistor and a 5 mH inductor (Fig. 9.3b). Thus, $\alpha = 1/2RC = 125,000\text{ s}^{-1}$, $\omega_0 = 1/\sqrt{LC} = 100,000\text{ rad/s}$, $s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} = -50,000\text{ s}^{-1}$ and $s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2} = -200,000\text{ s}^{-1}$.

► **Devise a plan.**

Since $\alpha > \omega_0$, the circuit is overdamped and so we expect a capacitor voltage of the form

$$v_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

We know s_1 and s_2 ; we need to obtain and invoke two initial conditions to determine A_1 and A_2 . To do this, we will analyze the circuit at $t = 0^-$ (Fig. 9.4a) to find $i_L(0^-)$ and $v_C(0^-)$. We will then analyze the circuit at $t = 0^+$ with the assumption that neither value changes.

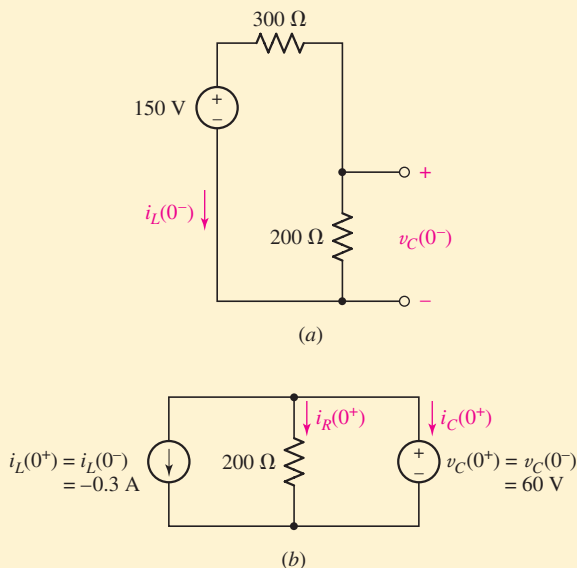
► **Construct an appropriate set of equations.**

From Fig. 9.4a, in which the inductor has been replaced with a short circuit and the capacitor with an open circuit, we see that

$$i_L(0^-) = -\frac{150}{200 + 300} = -300\text{ mA}$$

and

$$v_C(0^-) = 150 \frac{200}{200 + 300} = 60\text{ V}$$



■ **FIGURE 9.4** (a) The equivalent circuit at $t = 0^-$; (b) equivalent circuit at $t = 0^+$, drawn using ideal sources to represent the initial inductor current and initial capacitor voltage.

In Fig. 9.4b, we draw the circuit at $t = 0^+$, representing the inductor current and capacitor voltage by ideal sources for simplicity. Since neither can change in zero time, we know that $v_C(0^+) = 60$ V.

► **Determine if additional information is required.**

We have an equation for the capacitor voltage: $v_C(t) = A_1 e^{-50,000t} + A_2 e^{-200,000t}$. We now know $v_C(0) = 60$ V, but a third equation is still required. Differentiating our capacitor voltage equation, we find

$$\frac{dv_C}{dt} = -50,000A_1 e^{-50,000t} - 200,000A_2 e^{-200,000t}$$

which can be related to the capacitor current as $i_C = C(dv_C/dt)$.

Returning to Fig. 9.4b, KCL yields

$$i_C(0^+) = -i_L(0^+) - i_R(0^+) = 0.3 - [v_C(0^+)/200] = 0$$

► **Attempt a solution.**

Application of our first initial condition yields

$$v_C(0) = A_1 + A_2 = 60$$

and application of our second initial condition yields

$$i_C(0) = -20 \times 10^{-9}(50,000A_1 + 200,000A_2) = 0$$

Solving, $A_1 = 80$ V and $A_2 = -20$ V, so that

$$v_C(t) = 80e^{-50,000t} - 20e^{-200,000t} \text{ V}, \quad t > 0$$

► **Verify the solution. Is it reasonable or expected?**

At the very least, we can check our solution at $t = 0$, verifying that $v_C(0) = 60$ V. Differentiating and multiplying by 20×10^{-9} , we can also verify that $i_C(0) = 0$. Also, since we have a source-free circuit for $t > 0$, we expect that $v_C(t)$ must eventually decay to zero as t approaches ∞ , which our solution does.

PRACTICE

9.2 After being open for a long time, the switch in Fig. 9.5 closes at $t = 0$. Find (a) $i_L(0^-)$; (b) $v_C(0^-)$; (c) $i_R(0^+)$; (d) $i_C(0^+)$; (e) $v_C(0.2)$.

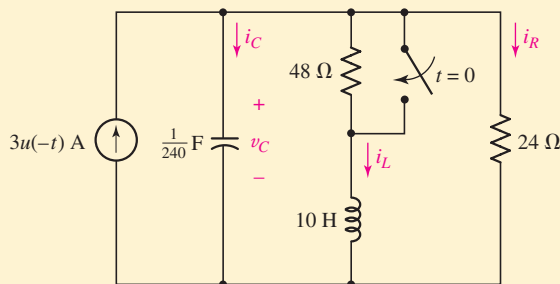


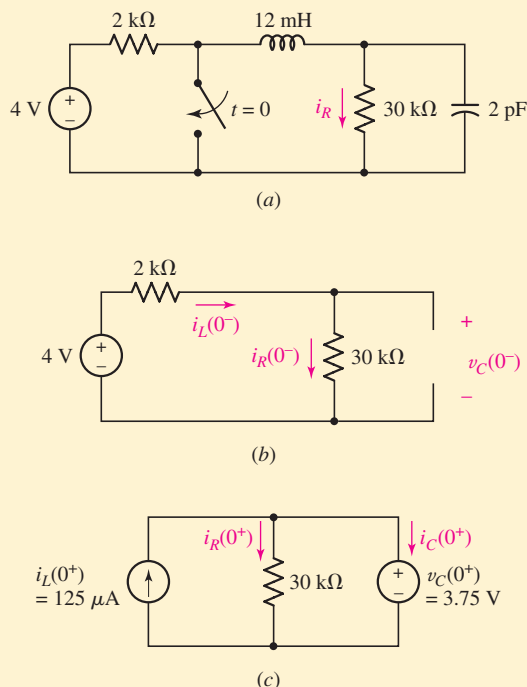
FIGURE 9.5

Ans: 1 A; 48 V; 2 A; -3 A; -17.54 V.

As noted previously, the form of the overdamped response applies to any voltage or current quantity, as we explore in the following example.

EXAMPLE 9.3

The circuit of Fig. 9.6a reduces to a simple parallel RLC circuit after $t = 0$. Determine an expression for the resistor current i_R valid for all time.



■ FIGURE 9.6 (a) Circuit for which i_R is required. (b) Equivalent circuit for $t = 0^-$. (c) Equivalent circuit for $t = 0^+$.

For $t > 0$, we have a parallel RLC circuit with $R = 30 \text{ k}\Omega$, $L = 12 \text{ mH}$, and $C = 2 \text{ pF}$. Thus, $\alpha = 8.333 \times 10^6 \text{ s}^{-1}$ and $\omega_0 = 6.455 \times 10^6 \text{ rad/s}$. We therefore expect an overdamped response, with $s_1 = -3.063 \times 10^6 \text{ s}^{-1}$ and $s_2 = -13.60 \times 10^6 \text{ s}^{-1}$, so that

$$i_R(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}, \quad t > 0 \quad [18]$$

To determine numerical values for A_1 and A_2 , we first analyze the circuit at $t = 0^-$, as drawn in Fig. 9.6b. We see that $i_L(0^-) = i_R(0^-) = 4/32 \times 10^3 = 125 \mu\text{A}$, and $v_C(0^-) = 4 \times 30/32 = 3.75 \text{ V}$.

In drawing the circuit at $t = 0^+$ (Fig. 9.6c), we only know that $i_L(0^+) = 125 \mu\text{A}$ and $v_C(0^+) = 3.75 \text{ V}$. However, by Ohm's law we can calculate that $i_R(0^+) = 3.75/30 \times 10^3 = 125 \mu\text{A}$, our first initial condition. Thus,

$$i_R(0) = A_1 + A_2 = 125 \times 10^{-6} \quad [19]$$

How do we obtain a *second* initial condition? If we multiply Eq. [18] by 30×10^3 , we obtain an expression for $v_C(t)$. Taking the derivative and multiplying by 2 pF yield an expression for $i_C(t)$:

$$i_C = C \frac{dv_C}{dt} = (2 \times 10^{-12})(30 \times 10^3)(A_1 s_1 e^{s_1 t} + A_2 s_2 e^{s_2 t})$$

By KCL,

$$i_C(0^+) = i_L(0^+) - i_R(0^+) = 0$$

Thus,

$$-(2 \times 10^{-12})(30 \times 10^3)(3.063 \times 10^6 A_1 + 13.60 \times 10^6 A_2) = 0 \quad [20]$$

Solving Eqs. [19] and [20], we find that $A_1 = 161.3 \mu\text{A}$ and $A_2 = -36.34 \mu\text{A}$. Thus,

$$i_R = \begin{cases} 125 \mu\text{A} & t < 0 \\ 161.3e^{-3.063 \times 10^6 t} - 36.34e^{-13.6 \times 10^6 t} \mu\text{A} & t > 0 \end{cases}$$

PRACTICE

9.3 Determine the current i_R through the resistor of Fig. 9.7 for $t > 0$ if $i_L(0^-) = 6 \text{ A}$ and $v_C(0^+) = 0 \text{ V}$. The configuration of the circuit prior to $t = 0$ is not known.

Ans: $i_R(t) = 2.437(e^{-7.823 \times 10^{10} t} - e^{-0.511 \times 10^{10} t}) \text{ A}$.

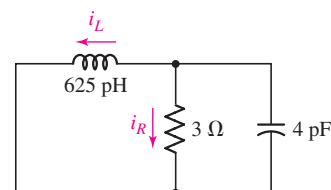


FIGURE 9.7 Circuit for Practice Problem 9.3.

Graphical Representation of the Overdamped Response

Now let us return to Eq. [17] and see what additional information we can determine about this circuit. We may interpret the first exponential term as having a time constant of 1 s and the other exponential, a time constant of $\frac{1}{6}$ s. Each starts with unity amplitude, but the latter decays more rapidly; $v(t)$ is never negative. As time becomes infinite, each term approaches zero, and the response itself dies out as it should. We therefore have a response curve which is zero at $t = 0$, is zero at $t = \infty$, and is never negative; since it is not everywhere zero, it must possess at least one maximum, and this is not a difficult point to determine exactly. We differentiate the response

$$\frac{dv}{dt} = 84(-e^{-t} + 6e^{-6t})$$

set the derivative equal to zero to determine the time t_m at which the voltage becomes maximum,

$$0 = -e^{-t_m} + 6e^{-6t_m}$$

manipulate once,

$$e^{5t_m} = 6$$

and obtain

$$t_m = 0.358 \text{ s}$$

and

$$v(t_m) = 48.9 \text{ V}$$

A reasonable sketch of the response may be made by plotting the two exponential terms $84e^{-t}$ and $84e^{-6t}$ and then taking their difference. This technique is illustrated by the curves of Fig. 9.8; the two exponentials are shown lightly, and their difference, the total response $v(t)$, is drawn as a colored line. The curves also verify our previous prediction that the functional behavior of $v(t)$ for very large t is $84e^{-t}$, the exponential term containing the smaller magnitude of s_1 and s_2 .

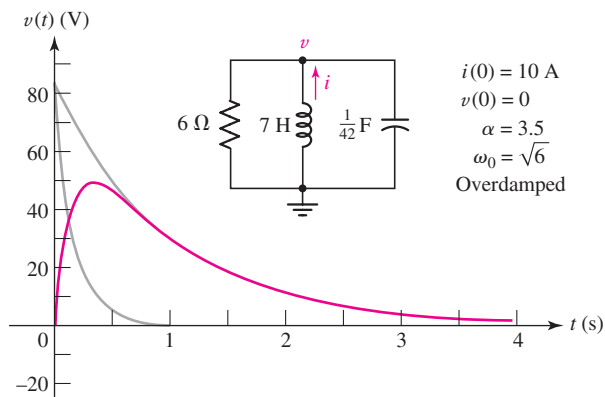


FIGURE 9.8 The response $v(t) = 84(e^{-t} - e^{-6t})$ of the network shown in Fig. 9.2.

A frequently asked question is the length of time it actually takes for the transient part of the response to disappear (or “*damp out*”). In practice, it is often desirable to have this transient response approach zero as rapidly as possible, that is, to minimize the **settling time** t_s . Theoretically, of course, t_s is infinite, because $v(t)$ never settles to zero in a finite time. However, a negligible response is present after the magnitude of $v(t)$ has settled to values that remain less than 1 percent of its maximum absolute value $|v_m|$. The time that is required for this to occur we define as the settling time. Since $|v_m| = v_m = 48.9 \text{ V}$ for our example, the settling time is the time required for the response to drop to 0.489 V. Substituting this value for $v(t)$ in Eq. [17] and neglecting the second exponential term, known to be negligible here, the settling time is found to be 5.15 s.



EXAMPLE 9.4

For $t > 0$, the capacitor current of a certain source-free parallel RLC circuit is given by $i_C(t) = 2e^{-2t} - 4e^{-t} \text{ A}$. Sketch the current in the range $0 < t < 5 \text{ s}$, and determine the settling time.

We first sketch the two terms as shown in Fig. 9.9, then subtract them to find $i_C(t)$. The maximum value is clearly $|-2| = 2 \text{ A}$. We therefore need to find the time at which $|i_C|$ has decreased to 20 mA, or

$$2e^{-2t_s} - 4e^{-t_s} = -0.02 \quad [21]$$

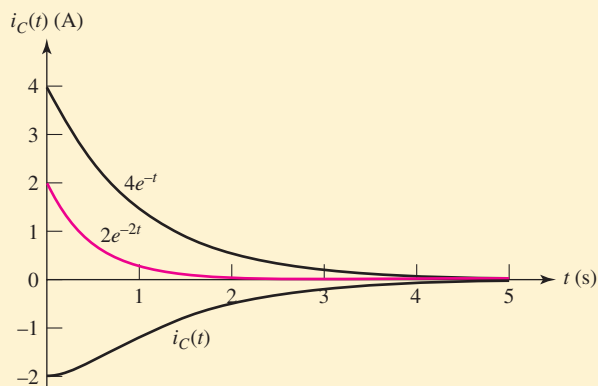


FIGURE 9.9 The current response $i_C(t) = 2e^{-2t} - 4e^{-t}$ A, sketched alongside its two components.

This equation can be solved using an iterative solver routine on a scientific calculator, which returns the solution $t_s = 5.296$ s. If such an option is not available, however, we can approximate Eq. [21] for $t \geq t_s$ as

$$-4e^{-t_s} = -0.02 \quad [22]$$

Solving,

$$t_s = -\ln\left(\frac{0.02}{4}\right) = 5.298 \text{ s} \quad [23]$$

which is reasonably close (better than 0.1% accuracy) to the exact solution.

PRACTICE

9.4 (a) Sketch the voltage $v_R(t) = 2e^{-t} - 4e^{-3t}$ V in the range $0 < t < 5$ s. (b) Estimate the settling time. (c) Calculate the maximum positive value and the time at which it occurs.

Ans: See Fig. 9.10; 5.9 s; 544 mV, 896 ms.

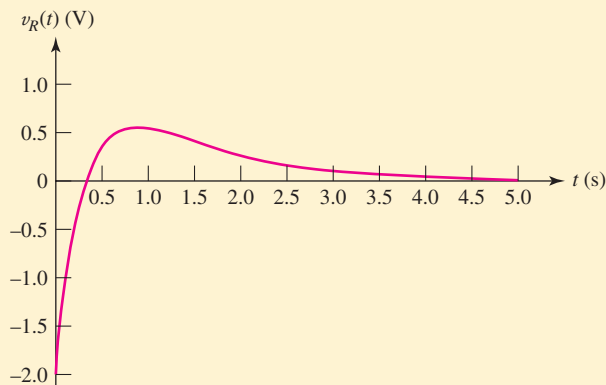


FIGURE 9.10 Response sketched for Practice Problem 9.4a.

9.3 CRITICAL DAMPING

The overdamped case is characterized by

$$\alpha > \omega_0$$

or

$$LC > 4R^2C^2$$

and leads to negative real values for s_1 and s_2 and to a response expressed as the algebraic sum of two negative exponentials.

Now let us adjust the element values until α and ω_0 are equal. This is a very special case which is termed **critical damping**. If we were to attempt to construct a parallel RLC circuit that is critically damped, we would be attempting an essentially impossible task, for we could never make α exactly equal to ω_0 . For completeness, however, we will discuss the critically damped circuit here, because it shows an interesting transition between overdamping and underdamping.

Critical damping is achieved when

$$\text{or} \quad \left. \begin{array}{l} \alpha = \omega_0 \\ LC = 4R^2C^2 \\ L = 4R^2C \end{array} \right\} \text{critical damping}$$

We can produce critical damping by changing the value of any of the three elements in the numerical example discussed at the end of Sec. 9.1. We will select R , increasing its value until critical damping is obtained, and thus leave ω_0 unchanged. The necessary value of R is $7\sqrt{6}/2 \, \Omega$; L is still 7 H, and C remains $\frac{1}{42}$ F. We thus find

$$\alpha = \omega_0 = \sqrt{6} \, \text{s}^{-1}$$

$$s_1 = s_2 = -\sqrt{6} \, \text{s}^{-1}$$

and recall the initial conditions that were specified, $v(0) = 0$ and $i(0) = 10$ A.

Form of a Critically Damped Response

We proceed to attempt to construct a response as the sum of two exponentials,

$$v(t) \stackrel{?}{=} A_1 e^{-\sqrt{6}t} + A_2 e^{-\sqrt{6}t}$$

which may be written as

$$v(t) \stackrel{?}{=} A_3 e^{-\sqrt{6}t}$$

At this point, some of us might be feeling that something's wrong. We have a response that contains only one arbitrary constant, but there are two initial conditions, $v(0) = 0$ and $i(0) = 10$ amperes, *both of which* must be satisfied by this single constant. If we select $A_3 = 0$, then $v(t) = 0$, which is consistent with our initial capacitor voltage. However, although there is no energy stored in the capacitor at $t = 0^+$, we have 350 J of energy initially stored in the inductor. This energy will lead to a transient current flowing out of the inductor, giving rise to a nonzero voltage across all three elements. This seems to be in direct conflict with our proposed solution.

"Impossible" is a pretty strong term. We make this statement because in practice it is unusual to obtain components that are closer than 1 percent of their specified values. Thus, obtaining L *precisely* equal to $4R^2C$ is theoretically possible, but not very likely, even if we're willing to measure a drawer full of components until we find the right ones.

If a mistake has not led to our difficulties, we must have begun with an incorrect assumption, and only one assumption has been made. We originally hypothesized that the differential equation could be solved by assuming an exponential solution, and this turns out to be incorrect for this single special case of critical damping. When $\alpha = \omega_0$, the differential equation, Eq. [4], becomes



$$\frac{d^2v}{dt^2} + 2\alpha \frac{dv}{dt} + \alpha^2 v = 0$$

The solution of this equation is not a tremendously difficult process, but we will avoid developing it here, since the equation is a standard type found in the usual differential-equation texts. The solution is

$$v = e^{-\alpha t} (A_1 t + A_2) \quad [24]$$

It should be noted that the solution is still expressed as the sum of two terms, where one term is the familiar negative exponential and the second is t times a negative exponential. We should also note that the solution contains the *two* expected arbitrary constants.

Finding Values for A_1 and A_2

Let us now complete our numerical example. After we substitute the known value of α in Eq. [24], obtaining

$$v = A_1 t e^{-\sqrt{6}t} + A_2 e^{-\sqrt{6}t}$$

we establish the values of A_1 and A_2 by first imposing the initial condition on $v(t)$ itself, $v(0) = 0$. Thus, $A_2 = 0$. This simple result occurs because the initial value of the response $v(t)$ was selected as zero; the more general case will require the solution of two equations simultaneously. The second initial condition must be applied to the derivative dv/dt just as in the overdamped case. We therefore differentiate, remembering that $A_2 = 0$:

$$\frac{dv}{dt} = A_1 t (-\sqrt{6}) e^{-\sqrt{6}t} + A_1 e^{-\sqrt{6}t}$$

evaluate at $t = 0$:

$$\left. \frac{dv}{dt} \right|_{t=0} = A_1$$

and express the derivative in terms of the initial capacitor current:

$$\left. \frac{dv}{dt} \right|_{t=0} = \frac{i_C(0)}{C} = \frac{i_R(0)}{C} + \frac{i(0)}{C}$$

where reference directions for i_C , i_R , and i are defined in Fig. 9.2. Thus,

$$A_1 = 420 \text{ V}$$

The response is, therefore,

$$v(t) = 420 t e^{-2.45t} \quad \text{V} \quad [25]$$

Graphical Representation of the Critically Damped Response

Before plotting this response in detail, let us again try to anticipate its form by qualitative reasoning. The specified initial value is zero, and Eq. [25] concurs. It is not immediately apparent that the response also approaches zero as t becomes infinitely large, because $te^{-2.45t}$ is an indeterminate form. However, this obstacle is easily overcome by use of L'Hôpital's rule, which yields

$$\lim_{t \rightarrow \infty} v(t) = 420 \lim_{t \rightarrow \infty} \frac{t}{e^{2.45t}} = 420 \lim_{t \rightarrow \infty} \frac{1}{2.45e^{2.45t}} = 0$$

and once again we have a response that begins and ends at zero and has positive values at all other times. A maximum value v_m again occurs at time t_m ; for our example,

$$t_m = 0.408 \text{ s} \quad \text{and} \quad v_m = 63.1 \text{ V}$$

This maximum is larger than that obtained in the overdamped case, and is a result of the smaller losses that occur in the larger resistor; the time of the maximum response is slightly later than it was with overdamping. The settling time may also be determined by solving

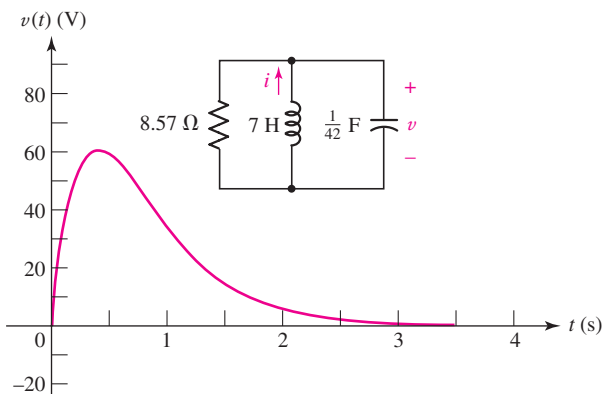
$$\frac{v_m}{100} = 420t_s e^{-2.45t_s}$$

for t_s (by trial-and-error methods or a calculator's SOLVE routine):

$$t_s = 3.12 \text{ s}$$



which is a considerably smaller value than that which arose in the overdamped case (5.15 s). As a matter of fact, it can be shown that, for given values of L and C , the selection of that value of R which provides critical damping will always give a shorter settling time than any choice of R that produces an overdamped response. However, a slight improvement (reduction) in settling time may be obtained by a further slight increase in resistance; a slightly underdamped response that will undershoot the zero axis before it dies out will yield the shortest settling time.



■ FIGURE 9.11 The response $v(t) = 420te^{-2.45t}$ of the network shown in Fig. 9.2 with R changed to provide critical damping.

The response curve for critical damping is drawn in Fig. 9.11; it may be compared with the overdamped (and underdamped) case by reference to Fig. 9.16.

EXAMPLE 9.5

Select a value for R_1 such that the circuit of Fig. 9.12 will be characterized by a critically damped response for $t > 0$, and a value for R_2 such that $v(0) = 2$ V.

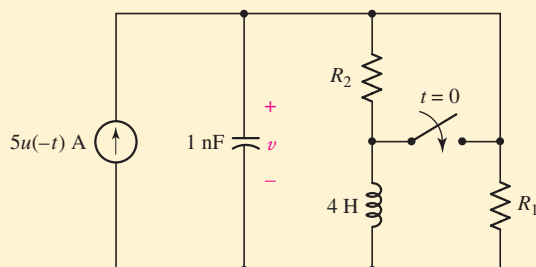


FIGURE 9.12 A circuit that reduces to a parallel RLC circuit after the switch is thrown.

We note that at $t = 0^-$, the current source is on, and the inductor can be treated as a short circuit. Thus, $v(0^-)$ appears across R_2 , and is given by

$$v(0^-) = 5R_2$$

and a value of $400 \text{ m}\Omega$ should be selected for R_2 to obtain $v(0) = 2$ V.

After the switch is thrown, the current source has turned itself off and R_2 is shorted. We are left with a parallel RLC circuit comprised of R_1 , a 4 H inductor, and a 1 nF capacitor.

We may now calculate (for $t > 0$)

$$\begin{aligned}\alpha &= \frac{1}{2RC} \\ &= \frac{1}{2 \times 10^{-9} R_1}\end{aligned}$$

and

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{4 \times 10^{-9}}} \\ &= 15,810 \text{ rad/s}\end{aligned}$$

Therefore, to establish a critically damped response in the circuit for $t > 0$, we need to set $R_1 = 31.63 \text{ k}\Omega$. (Note: since we have rounded to four significant figures, the pedantic can rightly argue that this is still not *exactly* a critically damped response—a difficult situation to create.)

PRACTICE

9.5 (a) Choose R_1 in the circuit of Fig. 9.13 so that the response after $t = 0$ will be critically damped. (b) Now select R_2 to obtain $v(0) = 100$ V. (c) Find $v(t)$ at $t = 1$ ms.

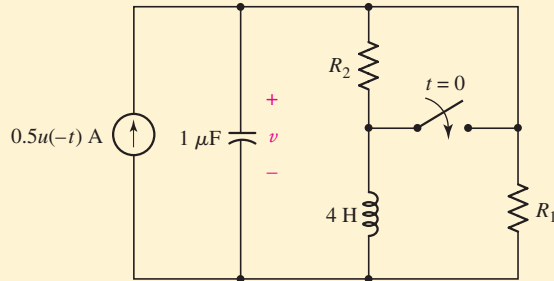


FIGURE 9.13

Ans: 1 k Ω ; 250 Ω ; -212 V.

9.4 THE UNDERDAMPED PARALLEL RLC CIRCUIT

Let us continue the process begun in Sec. 9.3 by increasing R once more to obtain what we will refer to as an *underdamped* response. Thus, the damping coefficient α decreases while ω_0 remains constant, α^2 becomes smaller than ω_0^2 , and the radicand appearing in the expressions for s_1 and s_2 becomes negative. This causes the response to take on a much different character, but it is fortunately not necessary to return to the basic differential equation again. By using complex numbers, the exponential response turns into a *damped sinusoidal response*; this response is composed entirely of real quantities, the complex quantities being necessary only for the derivation.¹

The Form of the Underdamped Response

We begin with the exponential form

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

where

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

and then let

$$\sqrt{\alpha^2 - \omega_0^2} = \sqrt{-1} \sqrt{\omega_0^2 - \alpha^2} = j \sqrt{\omega_0^2 - \alpha^2}$$

where $j \equiv \sqrt{-1}$.

We now take the new radical, which is real for the underdamped case, and call it ω_d , the *natural resonant frequency*:

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

The response may now be written as

$$v(t) = e^{-\alpha t} (A_1 e^{j\omega_d t} + A_2 e^{-j\omega_d t}) \quad [26]$$

Electrical engineers use “ j ” instead of “ i ” to represent $\sqrt{-1}$ to avoid confusion with currents.

(1) A review of complex numbers is presented in Appendix 5.

or, in the longer but equivalent form,

$$v(t) = e^{-\alpha t} \left\{ (A_1 + A_2) \left[\frac{e^{j\omega_d t} + e^{-j\omega_d t}}{2} \right] + j(A_1 - A_2) \left[\frac{e^{j\omega_d t} - e^{-j\omega_d t}}{j2} \right] \right\}$$

Applying identities described in Appendix 5, the term in the first square brackets in the preceding equation is identically equal to $\cos \omega_d t$, and the second is identically $\sin \omega_d t$. Hence,

$$v(t) = e^{-\alpha t} [(A_1 + A_2) \cos \omega_d t + j(A_1 - A_2) \sin \omega_d t]$$

and the multiplying factors may be assigned new symbols:

$$v(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t) \quad [27]$$

where Eqs. [26] and [27] are identical.

It may seem a little odd that our expression originally appeared to have a complex component, and now is purely real. However, we should remember that we originally allowed for A_1 and A_2 to be complex as well as s_1 and s_2 . In any event, if we are dealing with the underdamped case, we have now left complex numbers behind. This must be true since α , ω_d , and t are real quantities, so that $v(t)$ itself must be a real quantity (which might be presented on an oscilloscope, a voltmeter, or a sheet of graph paper). Equation [27] is the desired functional form for the underdamped response, and its validity may be checked by direct substitution in the original differential equation; this exercise is left to the doubters. The two real constants B_1 and B_2 are again selected to fit the given initial conditions.



We return to our simple parallel RLC circuit of Fig. 9.2 with $R = 6 \, \Omega$, $C = 1/42 \, \text{F}$, and $L = 7 \, \text{H}$, but now increase the resistance further to $10.5 \, \Omega$. Thus,

$$\alpha = \frac{1}{2RC} = 2 \, \text{s}^{-1}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \sqrt{6} \, \text{s}^{-1}$$

and

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{2} \, \text{rad/s}$$

Except for the evaluation of the arbitrary constants, the response is now known:

$$v(t) = e^{-2t} (B_1 \cos \sqrt{2}t + B_2 \sin \sqrt{2}t)$$

Finding Values for B_1 and B_2

The determination of the two constants proceeds as before. If we still assume that $v(0) = 0$ and $i(0) = 10$, then B_1 must be zero. Hence

$$v(t) = B_2 e^{-2t} \sin \sqrt{2}t$$

The derivative is

$$\frac{dv}{dt} = \sqrt{2} B_2 e^{-2t} \cos \sqrt{2}t - 2 B_2 e^{-2t} \sin \sqrt{2}t$$

and at $t = 0$ it becomes

$$\left. \frac{dv}{dt} \right|_{t=0} = \sqrt{2}B_2 = \frac{i_C(0)}{C} = 420$$

where i_C is defined in Fig. 9.2. Therefore,

$$v(t) = 210\sqrt{2}e^{-2t} \sin \sqrt{2}t$$

Graphical Representation of the Underdamped Response

Notice that, as before, this response function has an initial value of zero because of the initial voltage condition we imposed, and a final value of zero because the exponential term vanishes for large values of t . As t increases from zero through small positive values, $v(t)$ increases as $210\sqrt{2} \sin \sqrt{2}t$, because the exponential term remains essentially equal to unity. But, at some time t_m , the exponential function begins to decrease more rapidly than $\sin \sqrt{2}t$ is increasing; thus $v(t)$ reaches a maximum v_m and begins to decrease. We should note that t_m is not the value of t for which $\sin \sqrt{2}t$ is a maximum, but must occur somewhat before $\sin \sqrt{2}t$ reaches its maximum.



When $t = \pi/\sqrt{2}$, $v(t)$ is zero. Thus, in the interval $\pi/\sqrt{2} < t < \sqrt{2}\pi$, the response is negative, becoming zero again at $t = \sqrt{2}\pi$. Hence, $v(t)$ is an *oscillatory* function of time and crosses the time axis an infinite number of times at $t = n\pi/\sqrt{2}$, where n is any positive integer. In our example, however, the response is only slightly underdamped, and the exponential term causes the function to die out so rapidly that most of the zero crossings will not be evident in a sketch.

The oscillatory nature of the response becomes more noticeable as α decreases. If α is zero, which corresponds to an infinitely large resistance, then $v(t)$ is an undamped sinusoid that oscillates with constant amplitude. There is never a time at which $v(t)$ drops and stays below 1 percent of its maximum value; the settling time is therefore infinite. This is not perpetual motion; we have merely assumed an initial energy in the circuit and have not provided any means to dissipate this energy. It is transferred from its initial location in the inductor to the capacitor, then returns to the inductor, and so on, forever.

The Role of Finite Resistance

A finite R in the parallel RLC circuit acts as a kind of electrical transfer agent. Every time energy is transferred from L to C or from C to L , the agent exacts a commission. Before long, the agent has taken all the energy, wantonly dissipating every last joule. The L and C are left without a joule of their own, without voltage and without current. Actual parallel RLC circuits can be made to have effective values of R so large that a natural undamped sinusoidal response can be maintained for years without supplying any additional energy.

Returning to our specific numerical problem, differentiation locates the first maximum of $v(t)$,

$$v_{m_1} = 71.8 \text{ V} \quad \text{at} \quad t_{m_1} = 0.435 \text{ s}$$

the succeeding minimum,

$$v_{m_2} = -0.845 \text{ V} \quad \text{at} \quad t_{m_2} = 2.66 \text{ s}$$

and so on. The response curve is shown in Fig. 9.14. Additional response curves for increasingly more underdamped circuits are shown in Fig. 9.15.

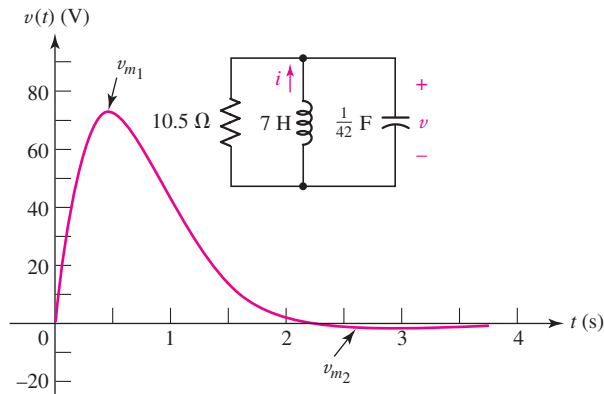


FIGURE 9.14 The response $v(t) = 210\sqrt{2}e^{-2t} \sin \sqrt{2}t$ of the network shown in Fig. 9.2 with R increased to produce an underdamped response.

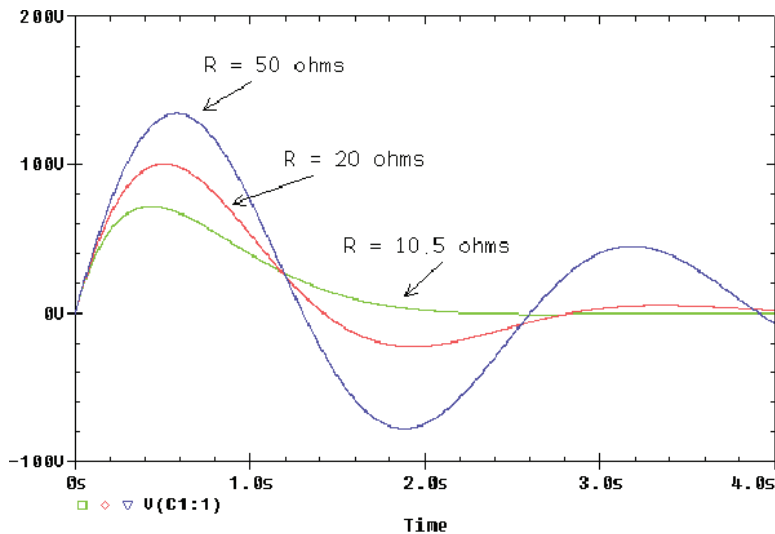


FIGURE 9.15 Simulated underdamped voltage response of the network for three different resistance values, showing an increase in the oscillatory behavior as R is increased.

The settling time may be obtained by a trial-and-error solution, and for $R = 10.5 \Omega$, it turns out to be 2.92 s, somewhat smaller than for critical damping. Note that t_s is greater than t_{m2} because the magnitude of v_{m2} is greater than 1 percent of the magnitude of v_{m1} . This suggests that a slight decrease in R would reduce the magnitude of the undershoot and permit t_s to be less than t_{m2} .

The overdamped, critically damped, and underdamped responses for this network as simulated by PSpice are shown on the same graph in Fig. 9.16. A comparison of the three curves makes the following general

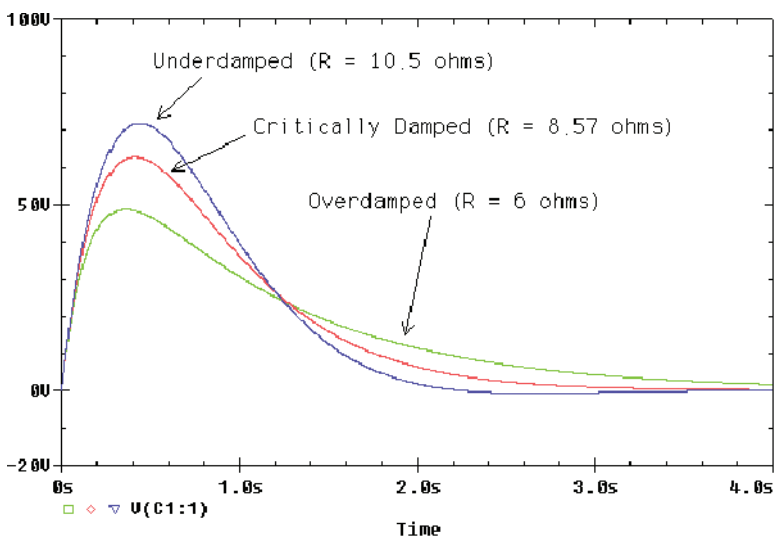


FIGURE 9.16 Simulated overdamped, critically damped, and underdamped voltage response for the example network, obtained by varying the value of the parallel resistance R .

conclusions plausible:

- When the damping is changed by increasing the size of the parallel resistance, the maximum magnitude of the response is greater and the amount of damping is smaller.
- The response becomes oscillatory when underdamping is present, and the minimum settling time is obtained for slight underdamping.

EXAMPLE 9.6

Determine $i_L(t)$ for the circuit of Fig. 9.17a, and plot the waveform.

At $t = 0$, both the 3 A source and the $48\ \Omega$ resistor are removed, leaving the circuit shown in Fig. 9.17b. Thus, $\alpha = 1.2\ \text{s}^{-1}$ and $\omega_0 = 4.899\ \text{rad/s}$. Since $\alpha < \omega_0$, the circuit is *underdamped*, and we therefore expect a response of the form

$$i_L(t) = e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t) \quad [28]$$

where $\omega_d = \sqrt{\omega_0^2 - \alpha^2} = 4.750\ \text{rad/s}$. The only remaining step is to find B_1 and B_2 .

Figure 9.17c shows the circuit as it exists at $t = 0^-$. We may replace the inductor with a short circuit and the capacitor with an open circuit; the result is $v_C(0^-) = 97.30\ \text{V}$ and $i_L(0^-) = 2.027\ \text{A}$. Since neither quantity can change in zero time, $v_C(0^+) = 97.30\ \text{V}$ and $i_L(0^+) = 2.027\ \text{A}$.

Substituting $i_L(0) = 2.027$ into Eq. [28] yields $B_1 = 2.027\ \text{A}$. To determine the other constant, we first differentiate Eq. [28]:

$$\frac{di_L}{dt} = e^{-\alpha t}(-B_1 \omega_d \sin \omega_d t + B_2 \omega_d \cos \omega_d t) - \alpha e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t) \quad [29]$$

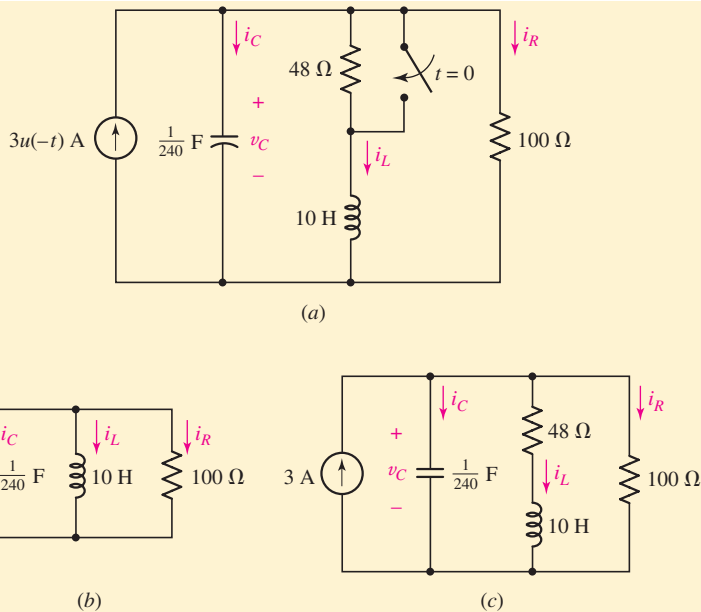


FIGURE 9.17 (a) A parallel RLC circuit for which the current $i_L(t)$ is desired. (b) Circuit for $t \geq 0$. (c) Circuit for determining the initial conditions.

and note that $v_L(t) = L(di_L/dt)$. Referring to the circuit of Fig. 9.17b, we see that $v_L(0^+) = v_C(0^+) = 97.3$ V. Thus, multiplying Eq. [29] by $L = 10$ H and setting $t = 0$, we find that

$$v_L(0) = 10(B_2\omega_d) - 10\alpha B_1 = 97.3$$

Solving, $B_2 = 2.561$ A, so that

$$i_L = e^{-1.2t}(2.027 \cos 4.75t + 2.561 \sin 4.75t) \quad \text{A}$$

which we have plotted in Fig. 9.18.

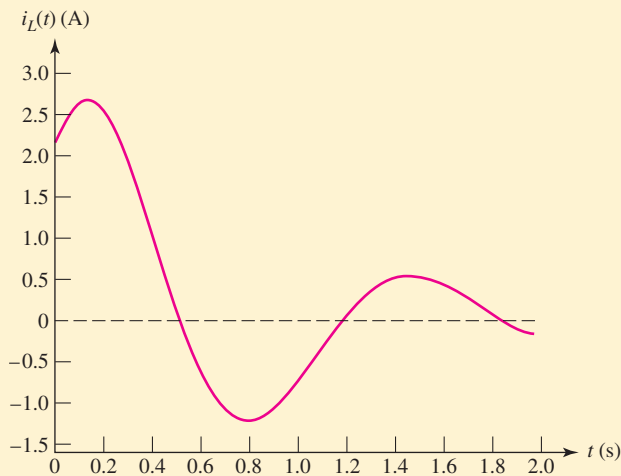


FIGURE 9.18 Plot of $i_L(t)$, showing obvious signs of being an underdamped response.

PRACTICE

9.6 The switch in the circuit of Fig. 9.19 has been in the left position for a long time; it is moved to the right at $t = 0$. Find (a) dv/dt at $t = 0^+$; (b) v at $t = 1$ ms; (c) t_0 , the first value of t greater than zero at which $v = 0$.

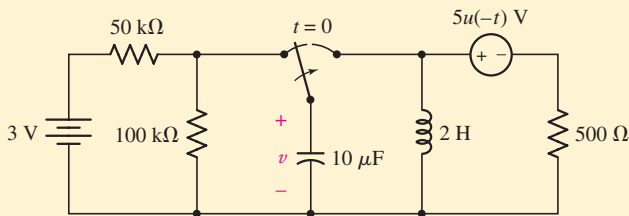


FIGURE 9.19

Ans: -1400 V/s; 0.695 V; 1.609 ms.

COMPUTER-AIDED ANALYSIS

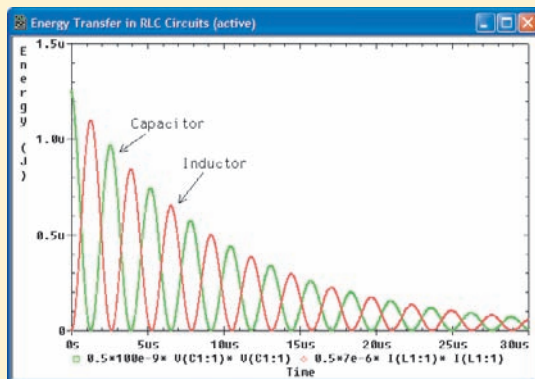
One useful feature in Probe is the ability to perform mathematical operations on the voltages and currents that result from a simulation. In this example, we will make use of that ability to show the transfer of energy in a parallel *RLC* circuit from a capacitor that initially stores a specific amount of energy ($1.25 \mu\text{J}$) to an inductor that initially stores no energy.

We choose a 100 nF capacitor and a $7 \mu\text{H}$ inductor, which immediately enables us to calculate $\omega_0 = 1.195 \times 10^6 \text{ s}^{-1}$. In order to consider overdamped, critically damped, and underdamped cases, we need to select the parallel resistance in such a way as to obtain $\alpha > \omega_0$ (*overdamped*), $\alpha = \omega_0$ (*critically damped*), and $\alpha < \omega_0$ (*underdamped*). From our previous discussions, we know that for a parallel *RLC* circuit $\alpha = (2RC)^{-1}$. We select $R = 4.1833 \Omega$ as a close approximation to the critically damped case; obtaining α precisely equal to ω_0 is effectively impossible. If we increase the resistance, the energy stored in the other two elements is dissipated more slowly, resulting in an underdamped response. We select $R = 100 \Omega$ so that we are well into this regime, and use $R = 1 \Omega$ (a very small resistance) to obtain an overdamped response.

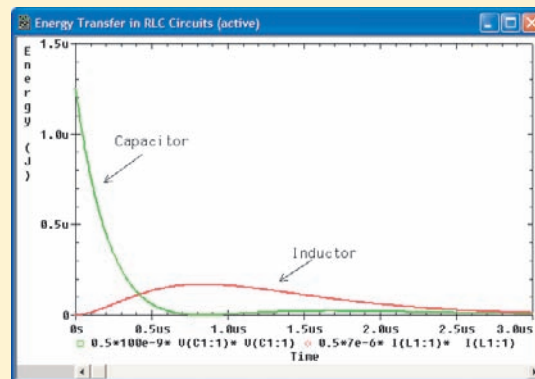
We therefore plan to run three separate simulations, varying only the resistance R between them. The $1.25 \mu\text{J}$ of energy initially stored in the capacitor equates to an initial voltage of 5 V , and so we set the initial condition of our capacitor accordingly.

Once Probe is launched, we select **Add** under the **Trace** menu. We wish to plot the energy stored in both the inductor and the capacitor as a function of time. For the capacitor, $w = \frac{1}{2}Cv^2$, so we click in the **Trace Expression** window, type in “ $0.5*100\text{E}-9*$ ” (without the quotes), click on $V(C1:1)$, return to the **Trace Expression** window and enter “ $*$ ”, click on $V(C1:1)$ once again, and then select **Ok**. We repeat the sequence to obtain the energy stored in the inductor, using $7\text{E}-6$ instead of $100\text{E}-9$, and clicking on $I(L1:1)$ instead of $V(C1:1)$.

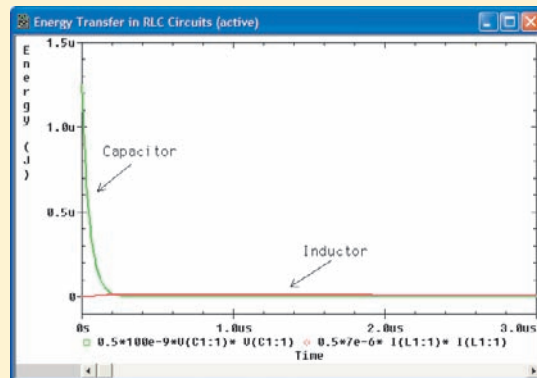
The Probe output plots for three separate simulations are provided in Fig. 9.20. In Fig. 9.20a, we see that the energy remaining in the circuit is continuously transferred back and forth between the capacitor and the inductor until it is (eventually) completely dissipated by the resistor. Decreasing the resistance to $4.1833\ \Omega$ yields a critically damped circuit, resulting in the energy plot of Fig. 9.20b. The oscillatory energy transfer between the capacitor and the inductor has been dramatically reduced. We see that the energy transferred to the inductor peaks at approximately $0.8\ \mu\text{s}$, and then drops to zero. The overdamped response is plotted in Fig. 9.20c. We note that the energy is dissipated much more quickly in the case of the overdamped response, and that very little energy is transferred to the inductor, since most of it is now quickly dissipated in the resistor.



(a)



(b)

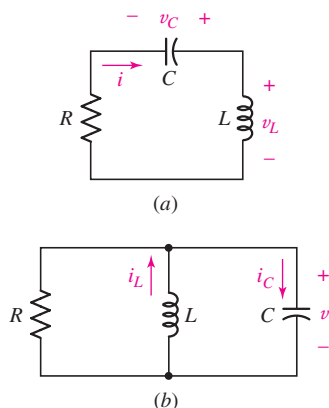


(c)

■ FIGURE 9.20 Energy transfer in a parallel *RLC* circuit with (a) $R = 100\ \Omega$ (underdamped); (b) $R = 4.1833\ \Omega$ (critically damped); and (c) $R = 1\ \Omega$ (overdamped).

9.5 THE SOURCE-FREE SERIES *RLC* CIRCUIT

We now wish to determine the natural response of a circuit model composed of an ideal resistor, an ideal inductor, and an ideal capacitor connected in *series*. The ideal resistor may represent a physical resistor connected into a series *LC* or *RLC* circuit; it may represent the ohmic losses and the losses in the ferromagnetic core of the inductor; or it may be used to represent all these and other energy-absorbing devices.



■ FIGURE 9.21 (a) The series RLC circuit which is the dual of (b) a parallel RLC circuit. Element values are, of course, not identical in the two circuits.

The series RLC circuit is the *dual* of the parallel RLC circuit, and this single fact is sufficient to make its analysis a trivial affair. Figure 9.21a shows the series circuit. The fundamental integrodifferential equation is

$$L \frac{di}{dt} + Ri + \frac{1}{C} \int_{t_0}^t i dt' - v_C(t_0) = 0$$

and should be compared with the analogous equation for the parallel RLC circuit, drawn again in Fig. 9.21b,

$$C \frac{dv}{dt} + \frac{1}{R} v + \frac{1}{L} \int_{t_0}^t v dt' - i_L(t_0) = 0$$

The respective second-order equations obtained by differentiating these two equations with respect to time are also duals:

$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{1}{C} i = 0 \quad [30]$$

$$C \frac{d^2 v}{dt^2} + \frac{1}{R} \frac{dv}{dt} + \frac{1}{L} v = 0 \quad [31]$$

Our complete discussion of the parallel RLC circuit is directly applicable to the series RLC circuit; the initial conditions on capacitor voltage and inductor current are equivalent to the initial conditions on inductor current and capacitor voltage; the *voltage* response becomes a *current* response. It is therefore possible to reread the previous four sections using dual language and thereby obtain a complete description of the series RLC circuit. This process, however, is apt to induce a mild neurosis after the first few paragraphs and does not really seem to be necessary.

A Brief Résumé of the Series Circuit Response

In terms of the circuit shown in Fig. 9.21a, the *overdamped response* is

$$i(t) = A_1 e^{s_{11}t} + A_2 e^{s_{21}t}$$

where

$$s_{1,2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

and thus

$$\alpha = \frac{R}{2L}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

The form of the *critically damped response* is

$$i(t) = e^{-\alpha t} (A_1 t + A_2)$$

and the *underdamped response* may be written

$$i(t) = e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$$

TABLE 9.1 Summary of Relevant Equations for Source-Free RLC Circuits

Type	Condition	Criteria	α	ω_0	Response
Parallel Series	Overdamped	$\alpha > \omega_0$	$\frac{1}{2RC}$ $\frac{R}{2L}$	$\frac{1}{\sqrt{LC}}$	$A_1 e^{s_1 t} + A_2 e^{s_2 t}$, where $s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$
Parallel Series	Critically damped	$\alpha = \omega_0$	$\frac{1}{2RC}$ $\frac{R}{2L}$	$\frac{1}{\sqrt{LC}}$	$e^{-\alpha t} (A_1 t + A_2)$
Parallel Series	Underdamped	$\alpha < \omega_0$	$\frac{1}{2RC}$ $\frac{R}{2L}$	$\frac{1}{\sqrt{LC}}$	$e^{-\alpha t} (B_1 \cos \omega_d t + B_2 \sin \omega_d t)$, where $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$

where

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

It is evident that if we work in terms of the parameters α , ω_0 , and ω_d , the mathematical forms of the responses for the dual situations are identical. An increase in α in either the series or parallel circuit, while keeping ω_0 constant, tends toward an overdamped response. The only caution that we need exert is in the computation of α , which is $1/2RC$ for the parallel circuit and $R/2L$ for the series circuit; thus, α is increased by increasing the series resistance or decreasing the parallel resistance. The key equations for parallel and series RLC circuits are summarized in Table 9.1 for convenience.



EXAMPLE 9.7

Given the series RLC circuit of Fig. 9.22 in which $L = 1$ H, $R = 2$ k Ω , $C = 1/401$ μ F, $i(0) = 2$ mA, and $v_C(0) = 2$ V, find and sketch $i(t)$, $t > 0$.

We find that $\alpha = R/2L = 1000$ s $^{-1}$ and $\omega_0 = 1/\sqrt{LC} = 20,025$ rad/s. This indicates an *underdamped* response; we therefore calculate the value of ω_d and obtain 20,000 rad/s. Except for the evaluation of the two arbitrary constants, the response is now known:

$$i(t) = e^{-1000t} (B_1 \cos 20,000t + B_2 \sin 20,000t)$$

Since we know that $i(0) = 2$ mA, we may substitute this value into our equation for $i(t)$ to obtain

$$B_1 = 0.002$$
 A

and thus

$$i(t) = e^{-1000t} (0.002 \cos 20,000t + B_2 \sin 20,000t) \quad \text{A}$$

(Continued on next page)

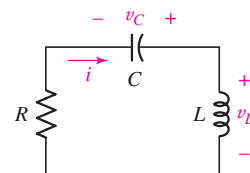


FIGURE 9.22 A simple source-free RLC circuit with energy stored in both the inductor and the capacitor at $t = 0$.

The remaining initial condition must be applied to the derivative; thus,

$$\frac{di}{dt} = e^{-1000t}(-40 \sin 20,000t + 20,000B_2 \cos 20,000t - 2 \cos 20,000t - 1000B_2 \sin 20,000t)$$

and

$$\begin{aligned} \left. \frac{di}{dt} \right|_{t=0} &= 20,000B_2 - 2 = \frac{v_L(0)}{L} \\ &= \frac{v_C(0) - Ri(0)}{L} \\ &= \frac{2 - 2000(0.002)}{1} = -2 \text{ A/s} \end{aligned}$$

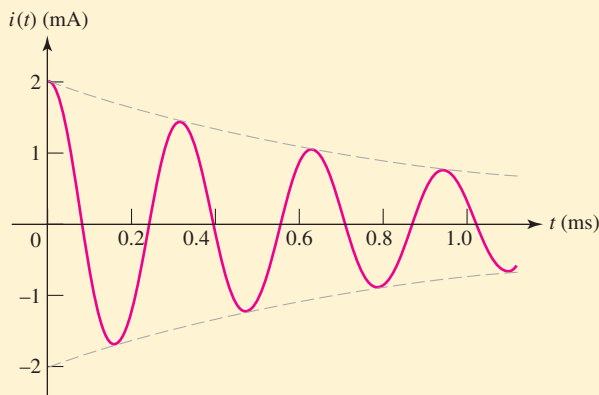
so that

$$B_2 = 0$$

The desired response is therefore

$$i(t) = 2e^{-1000t} \cos 20,000t \quad \text{mA}$$

A good sketch may be made by first drawing in the two portions of the exponential *envelope*, $2e^{-1000t}$ and $-2e^{-1000t}$ mA, as shown by the broken lines in Fig. 9.23. The location of the quarter-cycle points of the sinusoidal wave at $20,000t = 0, \pi/2, \pi$, etc., or $t = 0.07854k$ ms, $k = 0, 1, 2, \dots$, by light marks on the time axis then permits the oscillatory curve to be sketched in quickly.



■ FIGURE 9.23 The current response in an underdamped series RLC circuit for which $\alpha = 1000 \text{ s}^{-1}$, $\omega_0 = 20,000 \text{ s}^{-1}$, $i(0) = 2 \text{ mA}$, and $v_C(0) = 2 \text{ V}$. The graphical construction is simplified by drawing in the envelope, shown as a pair of broken lines.

The settling time can be determined easily here by using the upper portion of the envelope. That is, we set $2e^{-1000t_s}$ mA equal to 1 percent of its maximum value, 2 mA. Thus, $e^{-1000t_s} = 0.01$, and $t_s = 4.61$ ms is the approximate value that is usually used.

PRACTICE

9.7 With reference to the circuit shown in Fig. 9.24, find (a) α ; (b) ω_0 ; (c) $i(0^+)$; (d) $di/dt|_{t=0^+}$; (e) $i(12 \text{ ms})$.

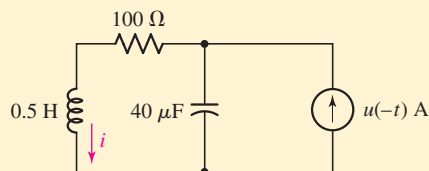


FIGURE 9.24

Ans: 100 s^{-1} ; 224 rad/s ; 1 A ; 0 ; -0.1204 A .

As a final example, we pause to consider situations where the circuit includes a dependent source. If no controlling current or voltage associated with the dependent source is of interest, we may simply find the Thévenin equivalent connected to the inductor and capacitor. Otherwise, we are likely faced with having to write an appropriate integrodifferential equation, take the indicated derivative, and solve the resulting differential equation as best we can.

EXAMPLE 9.8

Find an expression for $v_C(t)$ in the circuit of Fig. 9.25a, valid for $t > 0$.

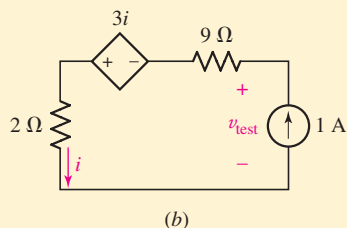
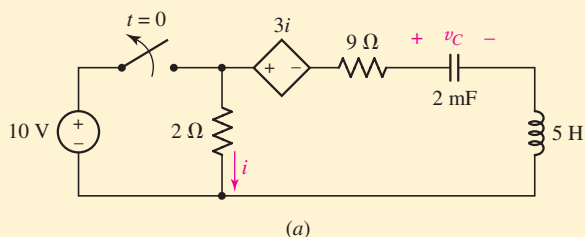


FIGURE 9.25 (a) An RLC circuit containing a dependent source.
(b) Circuit for finding R_{eq} .

As we are interested only in $v_C(t)$, it is perfectly acceptable to begin by finding the Thévenin equivalent resistance connected in series with the

(Continued on next page)

inductor and capacitor at $t = 0^+$. We do this by connecting a 1 A source as shown in Fig. 9.25b, from which we deduce that

$$v_{\text{test}} = 11i - 3i = 8i = 8(1) = 8 \text{ V}$$

Thus, $R_{\text{eq}} = 8 \Omega$, so $\alpha = R/2L = 0.8 \text{ s}^{-1}$ and $\omega_0 = 1/\sqrt{LC} = 10 \text{ rad/s}$, meaning that we expect an underdamped response with $\omega_d = 9.968 \text{ rad/s}$ and the form

$$v_C(t) = e^{-0.8t}(B_1 \cos 9.968t + B_2 \sin 9.968t) \quad [32]$$

In considering the circuit at $t = 0^-$, we note that $i_L(0^-) = 0$ due to the presence of the capacitor. By Ohm's law, $i(0^-) = 5 \text{ A}$, so

$$v_C(0^+) = v_C(0^-) = 10 - 3i = 10 - 15 = -5 \text{ V}$$

This last condition substituted into Eq. [32] yields $B_1 = -5 \text{ V}$. Taking the derivative of Eq. [32] and evaluating at $t = 0$ yield

$$\left. \frac{dv_C}{dt} \right|_{t=0} = -0.8B_1 + 9.968B_2 = 4 + 9.968B_2 \quad [33]$$

We see from Fig. 9.25a that

$$i = -C \frac{dv_C}{dt}$$

Thus, making use of the fact that $i(0^+) = i_L(0^-) = 0$ in Eq. [33] yields $B_2 = -0.4013 \text{ V}$, and we may write

$$v_C(t) = -e^{-0.8t}(5 \cos 9.968t + 0.4013 \sin 9.968t) \quad \text{V} \quad t > 0$$

The PSpice simulation of this circuit, shown in Fig. 9.26, confirms our analysis.

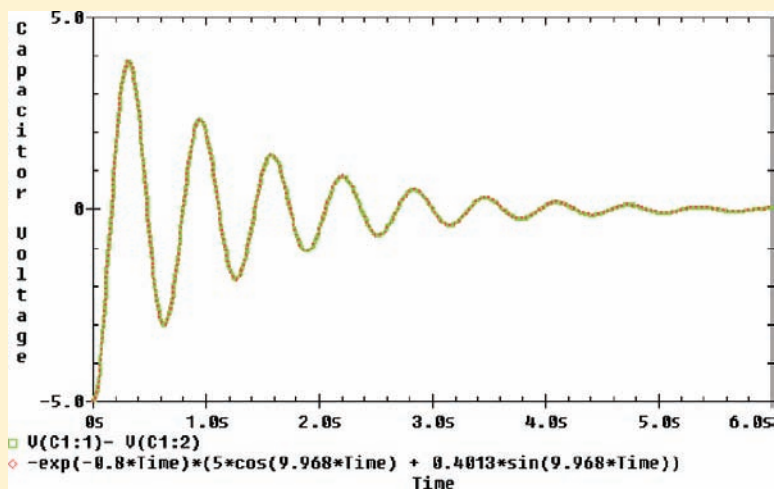


FIGURE 9.26 PSpice simulation of the circuit shown in Fig. 9.25a. The analytical result is plotted using a dashed red line.

PRACTICE

9.8 Find an expression for $i_L(t)$ in the circuit of Fig. 9.27, valid for $t > 0$, if $v_C(0^-) = 10$ V and $i_L(0^-) = 0$. *Note that although it is not helpful to apply Thévenin techniques in this instance, the action of the dependent source links v_C and i_L such that a first-order linear differential equation results.*

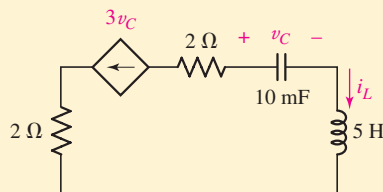


FIGURE 9.27 Circuit for Practice Problem 9.8.

Ans: $i_L(t) = -30e^{-300t}$ A, $t > 0$.

IMPORTANT

9.6 THE COMPLETE RESPONSE OF THE *RLC* CIRCUIT

We now consider those *RLC* circuits in which dc sources are switched into the network and produce forced responses that do not necessarily vanish as time becomes infinite.

The general solution is obtained by the same procedure that was followed for *RL* and *RC* circuits. The basic steps are (not necessarily in this order) as follows:

1. Determine the initial conditions.
2. Obtain a numerical value for the forced response.
3. Write the appropriate form of the natural response with the necessary number of arbitrary constants.
4. Add the forced response and natural response to form the complete response.
5. Evaluate the response and its derivative at $t = 0$, and employ the initial conditions to solve for the values of the unknown constants.

We note that *it is generally this last step that causes the most trouble for students*, as the circuit must be carefully evaluated at $t = 0$ to make full use of the initial conditions. Consequently, although the determination of the initial conditions is basically no different for a circuit containing dc sources from what it is for the source-free circuits that we have already covered in some detail, this topic will receive particular emphasis in the examples that follow.



Most of the confusion in determining and applying the initial conditions arises for the simple reason that we do not have a rigorous set of rules laid down for us to follow. At some point in each analysis, a situation usually arises in which some thinking is involved that is more or less unique to that particular problem. This is almost always the source of the difficulty.

The Easy Part

The *complete* response (arbitrarily assumed to be a voltage response) of a second-order system consists of a *forced* response,

$$v_f(t) = V_f$$

which is a constant for dc excitation, and a *natural* response,

$$v_n(t) = Ae^{s_1 t} + Be^{s_2 t}$$

Thus,

$$v(t) = V_f + Ae^{s_1 t} + Be^{s_2 t}$$

We assume that s_1 , s_2 , and V_f have already been determined from the circuit and the given forcing functions; A and B remain to be found. The last equation shows the functional interdependence of A , B , v , and t ; and substitution of the known value of v at $t = 0^+$ thus provides us with a single equation relating A and B , $v(0^+) = V_f + A + B$. *This is the easy part.*

The Other Part

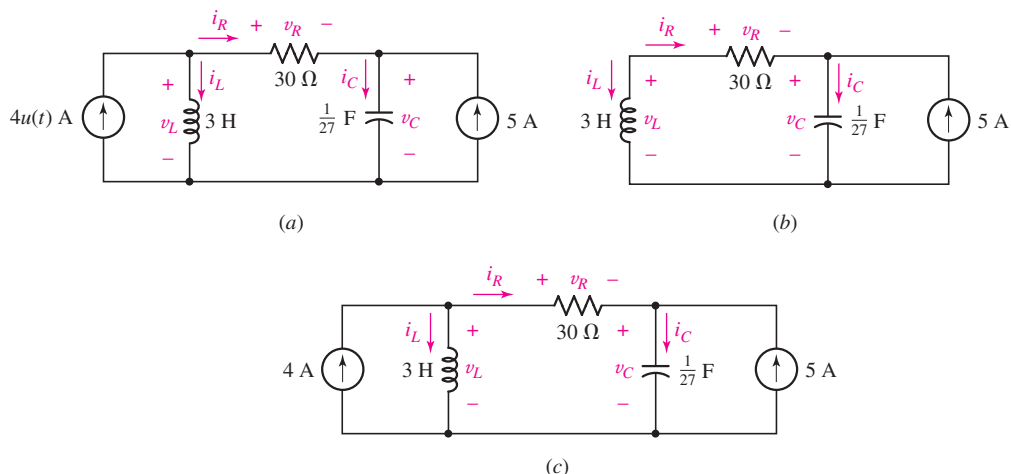
Another relationship between A and B is necessary, unfortunately, and this is normally obtained by taking the derivative of the response,

$$\frac{dv}{dt} = 0 + s_1 Ae^{s_1 t} + s_2 Be^{s_2 t}$$

and inserting the known value of dv/dt at $t = 0^+$. We thus have two equations relating A and B , and these may be solved simultaneously to evaluate the two constants.

The only remaining problem is that of determining the values of v and dv/dt at $t = 0^+$. Let us suppose that v is a capacitor voltage, v_C . Since $i_C = C dv_C/dt$, we should recognize the relationship between the initial value of dv/dt and the initial value of some capacitor current. If we can establish a value for this initial capacitor current, then we will automatically establish the value of dv/dt . Students are usually able to get $v(0^+)$ very easily, but are inclined to stumble a bit in finding the initial value of dv/dt . If we had selected an inductor current i_L as our response, then the initial value of di_L/dt would be intimately related to the initial value of some inductor voltage. Variables other than capacitor voltages and inductor currents are determined by expressing their initial values and the initial values of their derivatives in terms of the corresponding values for v_C and i_L .

We will illustrate the procedure and find all these values by the careful analysis of the circuit shown in Fig. 9.28. To simplify the analysis, an unusual value of capacitance is used again.



■ FIGURE 9.28 (a) An RLC circuit that is used to illustrate several procedures by which the initial conditions may be obtained. The desired response is nominally taken to be $v_C(t)$. (b) $t = 0^-$. (c) $t > 0$.

EXAMPLE 9.9

There are three passive elements in the circuit shown in Fig. 9.28a, and a voltage and a current are defined for each. Find the values of these six quantities at both $t = 0^-$ and $t = 0^+$.

Our object is to find the value of each current and voltage at both $t = 0^-$ and $t = 0^+$. Once these quantities are known, the initial values of the derivatives may be found easily.

1. $t = 0^-$ At $t = 0^-$, only the right-hand current source is active as depicted in Fig. 9.28b. The circuit is assumed to have been in this state forever, so all currents and voltages are constant. Thus, a dc current through the inductor requires zero voltage across it:

$$v_L(0^-) = 0$$

and a dc voltage across the capacitor ($-v_R$) requires zero current through it:

$$i_C(0^-) = 0$$

We next apply Kirchhoff's current law to the right-hand node to obtain

$$i_R(0^-) = -5 \text{ A}$$

which also yields

$$v_R(0^-) = -150 \text{ V}$$

We may now use Kirchhoff's voltage law around the left-hand mesh, finding

$$v_C(0^-) = 150 \text{ V}$$

while KCL enables us to find the inductor current,

$$i_L(0^-) = 5 \text{ A}$$

2. $t = 0^+$ During the interval from $t = 0^-$ to $t = 0^+$, the left-hand current source becomes active and many of the voltage and current values at $t = 0^-$ will change abruptly. The corresponding circuit is shown in Fig. 9.28c. However, we should *begin by focusing our attention on those quantities which cannot change, namely, the inductor current and the capacitor voltage*. Both of these must remain constant during the switching interval. Thus,

$$i_L(0^+) = 5 \text{ A} \quad \text{and} \quad v_C(0^+) = 150 \text{ V}$$

Since two currents are now known at the left node, we next obtain

$$i_R(0^+) = -1 \text{ A} \quad \text{and} \quad v_R(0^+) = -30 \text{ V}$$

so that

$$i_C(0^+) = 4 \text{ A} \quad \text{and} \quad v_L(0^+) = 120 \text{ V}$$

and we have our six initial values at $t = 0^-$ and six more at $t = 0^+$.

Among these last six values, only the capacitor voltage and the inductor current are unchanged from the $t = 0^-$ values.

We could have employed a slightly different method to evaluate these currents and voltages at $t = 0^-$ and $t = 0^+$. Prior to the switching operation, only direct currents and voltages exist in the circuit. The inductor may therefore be replaced by a short circuit, its dc equivalent, while the capacitor is replaced by an open circuit. Redrawn in this manner, the circuit of Fig. 9.28a appears as shown in Fig. 9.29a. Only the current source at the right is active, and its 5 A flow through the resistor and the inductor. We therefore have $i_R(0^-) = -5$ A and $v_R(0^-) = -150$ V, $i_L(0^-) = 5$ A and $v_L(0^-) = 0$, and $i_C(0^-) = 0$ and $v_C(0^-) = 150$ V, as before.

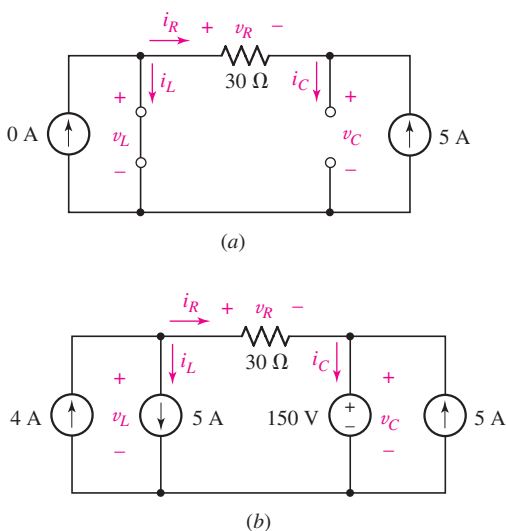


FIGURE 9.29 (a) A simple circuit equivalent to the circuit of Fig. 9.28a for $t = 0^-$. (b) Equivalent circuit with labeled voltages and currents valid at the instant defined by $t = 0^+$.

We now turn to the problem of drawing an equivalent circuit that will assist us in determining the several voltages and currents at $t = 0^+$. *Each capacitor voltage and each inductor current must remain constant during the switching interval.* These conditions are ensured by replacing the inductor with a current source and the capacitor with a voltage source. Each source serves to maintain a constant response during the discontinuity. The equivalent circuit of Fig. 9.29b results. It should be noted that the circuit shown in Fig. 9.29b is valid *only for the interval between 0^- and 0^+* .

The voltages and currents at $t = 0^+$ are obtained by analyzing this dc circuit. The solution is not difficult, but the relatively large number of sources present in the network does produce a somewhat strange sight. However, problems of this type were solved in Chap. 3, and nothing new is involved. Attacking the currents first, we begin at the upper left node and see that $i_R(0^+) = 4 - 5 = -1$ A. Moving to the upper right node, we find that $i_C(0^+) = -1 + 5 = 4$ A. And, of course, $i_L(0^+) = 5$ A.

Next we consider the voltages. Using Ohm's law, we see that $v_R(0^+) = 30(-1) = -30$ V. For the inductor, KVL gives us $v_L(0^+) = -30 + 150 = 120$ V. Finally, including $v_C(0^+) = 150$ V, we have all the values at $t = 0^+$.

PRACTICE

9.9 Let $i_s = 10u(-t) - 20u(t)$ A in Fig. 9.30. Find (a) $i_L(0^-)$; (b) $v_C(0^+)$; (c) $v_R(0^+)$; (d) $i_L(\infty)$; (e) $i_L(0.1 \text{ ms})$.

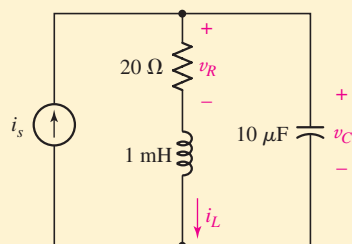


FIGURE 9.30

Ans: 10 A; 200 V; 200 V; -20 A; 2.07 A.

EXAMPLE 9.10

Complete the determination of the initial conditions in the circuit of Fig. 9.28, repeated in Fig. 9.31, by finding values at $t = 0^+$ for the first derivatives of the three voltage and three current variables defined on the circuit diagram.

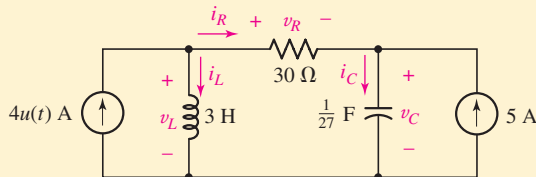


FIGURE 9.31 Circuit of Fig. 9.28, repeated for Example 9.10.

We begin with the two energy storage elements. For the inductor,

$$v_L = L \frac{di_L}{dt}$$

and, specifically,

$$v_L(0^+) = L \left. \frac{di_L}{dt} \right|_{t=0^+}$$

Thus,

$$\left. \frac{di_L}{dt} \right|_{t=0^+} = \frac{v_L(0^+)}{L} = \frac{120}{3} = 40 \text{ A/s}$$

Similarly,

$$\left. \frac{dv_C}{dt} \right|_{t=0^+} = \frac{i_C(0^+)}{C} = \frac{4}{1/27} = 108 \text{ V/s}$$

(Continued on next page)

The other four derivatives may be determined by realizing that KCL and KVL are both satisfied by the derivatives also. For example, at the left-hand node in Fig. 9.31,

$$4 - i_L - i_R = 0 \quad t > 0$$

and thus,

$$0 - \frac{di_L}{dt} - \frac{di_R}{dt} = 0 \quad t > 0$$

and therefore,

$$\left. \frac{di_R}{dt} \right|_{t=0^+} = -40 \text{ A/s}$$

The three remaining initial values of the derivatives are found to be

$$\left. \frac{dv_R}{dt} \right|_{t=0^+} = -1200 \text{ V/s}$$

$$\left. \frac{dv_L}{dt} \right|_{t=0^+} = -1092 \text{ V/s}$$

and

$$\left. \frac{di_C}{dt} \right|_{t=0^+} = -40 \text{ A/s}$$

Before leaving this problem of the determination of the necessary initial values, we should point out that at least one other powerful method of determining them has been omitted: we could have written general nodal or loop equations for the original circuit. Then the substitution of the known zero values of inductor voltage and capacitor current at $t = 0^-$ would uncover several other response values at $t = 0^-$ and enable the remainder to be found easily. A similar analysis at $t = 0^+$ must then be made. This is an important method, and it becomes a necessary one in more complicated circuits which cannot be analyzed by our simpler step-by-step procedures.

Now let us briefly complete the determination of the response $v_C(t)$ for the original circuit of Fig. 9.31. With both sources dead, the circuit appears as a series *RLC* circuit and s_1 and s_2 are easily found to be -1 and -9 , respectively. The forced response may be found by inspection or, if necessary, by drawing the dc equivalent, which is similar to Fig. 9.29a, with the addition of a 4 A current source. The forced response is 150 V. Thus,

$$v_C(t) = 150 + Ae^{-t} + Be^{-9t}$$

and

$$v_C(0^+) = 150 = 150 + A + B$$

or

$$A + B = 0$$

Then,

$$\frac{dv_C}{dt} = -Ae^{-t} - 9Be^{-9t}$$

and

$$\left. \frac{dv_C}{dt} \right|_{t=0^+} = 108 = -A - 9B$$

Finally,

$$A = 13.5 \quad B = -13.5$$

and

$$v_C(t) = 150 + 13.5(e^{-t} - e^{-9t}) \quad \text{V}$$

A Quick Summary of the Solution Process

In summary, then, whenever we wish to determine the transient behavior of a simple three-element *RLC* circuit, we must first decide whether we are confronted with a series or a parallel circuit, so that we may use the correct relationship for α . The two equations are

$$\alpha = \frac{1}{2RC} \quad (\text{parallel } RLC)$$

$$\alpha = \frac{R}{2L} \quad (\text{series } RLC)$$

Our second decision is made after comparing α with ω_0 , which is given for either circuit by

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

If $\alpha > \omega_0$, the circuit is *overdamped*, and the natural response has the form

$$f_n(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

where

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$

If $\alpha = \omega_0$, then the circuit is *critically damped* and

$$f_n(t) = e^{-\alpha t}(A_1 t + A_2)$$

And finally, if $\alpha < \omega_0$, then we are faced with the *underdamped* response,

$$f_n(t) = e^{-\alpha t}(A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

where

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

Our last decision depends on the independent sources. If there are none acting in the circuit after the switching or discontinuity is completed, then the circuit is source-free and the natural response accounts for the complete response. If independent sources are still present, then the circuit is driven and a forced response must be determined. The complete response is then the sum

$$f(t) = f_f(t) + f_n(t)$$

This is applicable to any current or voltage in the circuit. Our final step is to solve for unknown constants given the initial conditions.

PRACTICAL APPLICATION

Modeling Automotive Suspension Systems

Earlier, we alluded to the fact that the concepts investigated in this chapter actually extend beyond the analysis of electric circuits. In fact, the general form of the differential equations we have been working with appears in many fields—we need only learn how to “translate” new parameter terminology. For example, consider a simple automotive suspension, as shown in Fig. 9.32. The piston is not attached to the cylinder, but *is* attached to both the spring and the wheel. The moving parts therefore are the spring, the piston, and the wheel.

We will model this physical system by first determining the forces in play. Defining a position function $p(t)$ which describes where the piston lies within the cylinder, we may write F_S , the force on the spring, as

$$F_S = Kp(t)$$

where K is known as the spring constant and has units of lb/ft. The force on the wheel F_W is equal to the mass of the wheel times its acceleration, or

$$F_W = m \frac{d^2 p(t)}{dt^2}$$

where m is measured in $\text{lb} \cdot \text{s}^2/\text{ft}$. Last but not least is the force of friction F_f acting on the piston

$$F_f = \mu_f \frac{dp(t)}{dt}$$



■ FIGURE 9.32 Typical automotive suspension system.
© Transtock Inc./Alamy.

where μ_f is the coefficient of friction, with units of $\text{lb} \cdot \text{s}/\text{ft}$.

From our basic physics courses we know that all forces acting in our system must sum to zero, so that

$$m \frac{d^2 p(t)}{dt^2} + \mu_f \frac{dp(t)}{dt} + Kp(t) = 0 \quad [34]$$

This equation most likely had the potential to give us nightmares at one point in our academic career, but no longer. We compare Eq. [32] to Eqs. [30] and [31] and immediately see a distinct resemblance, at least in the general form. Choosing Eq. [30], the differential equation describing the inductor current of a series-connected *RLC* circuit, we observe the following correspondences:

Mass	m	→	inductance	L
Coefficient of friction	μ_f	→	resistance	R
Spring constant	K	→	inverse of the capacitance	C^{-1}
Position variable	$p(t)$	→	current variable	$i(t)$

So, if we are willing to talk about feet instead of amperes, $\text{lb} \cdot \text{s}^2/\text{ft}$ instead of H, ft/lb instead of F, and $\text{lb} \cdot \text{s}/\text{ft}$ instead of Ω , we can apply our newly found skills at modeling *RLC* circuits to the task of evaluating automotive shock absorbers.

Take a typical car wheel of 70 lb. The mass is found by dividing the weight by the earth’s gravitational acceleration (32.17 ft/s^2), resulting in $m = 2.176 \text{ lb} \cdot \text{s}^2/\text{ft}$. The curb weight of our car is 1985 lb, and the static displacement of the spring is 4 inches (no passengers). The spring constant is obtained by dividing the weight on each shock absorber by the static displacement, so that we have $K = (\frac{1}{4})(1985)(3 \text{ ft}^{-1}) = 1489 \text{ lb}/\text{ft}$. We are also told that the coefficient of friction for our piston-cylinder assembly is $65 \text{ lb} \cdot \text{s}/\text{ft}$. Thus, we can simulate our shock absorber by modeling it with a series *RLC* circuit having $R = 65 \Omega$, $L = 2.176 \text{ H}$, and $C = K^{-1} = 671.6 \mu\text{F}$.

The resonant frequency of our shock absorber is $\omega_0 = (LC)^{-1/2} = 26.16 \text{ rad/s}$, and the damping coefficient is $\alpha = R/2L = 14.94 \text{ s}^{-1}$. Since $\alpha < \omega_0$, our shock absorber represents an underdamped system; this means that we expect a bounce or two after we run over a pothole. A stiffer shock (larger coefficient of friction, or a larger resistance in our circuit model) is typically desirable when curves are taken at high speeds—at some point this corresponds to an overdamped response. However, if most of our driving is over unpaved roads, a slightly underdamped response is preferable.

PRACTICE

9.10 Let $v_s = 10 + 20u(t)$ V in the circuit of Fig. 9.33. Find (a) $i_L(0)$; (b) $v_C(0)$; (c) $i_{L,f}$; (d) $i_L(0.1 \text{ s})$.

Ans: 0.2 A; 10 V; 0.6 A; 0.319 A.

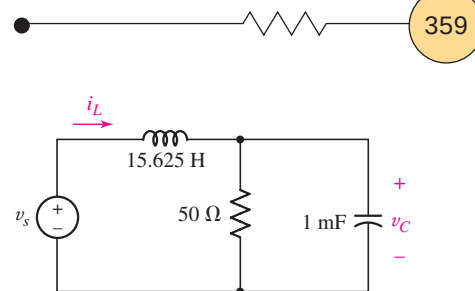


FIGURE 9.33

9.7 THE LOSSLESS LC CIRCUIT

When we considered the source-free RLC circuit, it became apparent that the resistor served to dissipate any initial energy stored in the circuit. At some point it might occur to us to ask what would happen if we could remove the resistor. If the value of the resistance in a parallel RLC circuit becomes infinite, or zero in the case of a series RLC circuit, we have a simple LC loop in which an oscillatory response can be maintained forever. Let us look briefly at an example of such a circuit, and then discuss another means of obtaining an identical response without the need of supplying any inductance.

Consider the source-free circuit of Fig. 9.34, in which the large values $L = 4 \text{ H}$ and $C = \frac{1}{36} \text{ F}$ are used so that the calculations will be simple. We let $i(0) = -\frac{1}{6} \text{ A}$ and $v(0) = 0$. We find that $\alpha = 0$ and $\omega_0^2 = 9 \text{ s}^{-2}$, so that $\omega_d = 3 \text{ rad/s}$. In the absence of exponential damping, the voltage v is simply

$$v = A \cos 3t + B \sin 3t$$

Since $v(0) = 0$, we see that $A = 0$. Next,

$$\left. \frac{dv}{dt} \right|_{t=0} = 3B = -\frac{i(0)}{1/36}$$

But $i(0) = -\frac{1}{6}$ ampere, and therefore $dv/dt = 6 \text{ V/s}$ at $t = 0$. We must have $B = 2 \text{ V}$ and so

$$v = 2 \sin 3t \quad \text{V}$$

which is an undamped sinusoidal response; in other words, our voltage response does not decay.

Now let us see how we might obtain this voltage without using an LC circuit. Our intentions are to write the differential equation that v satisfies and then to develop a configuration of op amps that will yield the solution of the equation. Although we are working with a specific example, the technique is a general one that can be used to solve any linear homogeneous differential equation.

For the LC circuit of Fig. 9.34, we select v as our variable and set the sum of the downward inductor and capacitor currents equal to zero:

$$\frac{1}{4} \int_{t_0}^t v \, dt' - \frac{1}{6} + \frac{1}{36} \frac{dv}{dt} = 0$$

Differentiating once, we have

$$\frac{1}{4}v + \frac{1}{36} \frac{d^2v}{dt^2} = 0$$

or

$$\frac{d^2v}{dt^2} = -9v$$

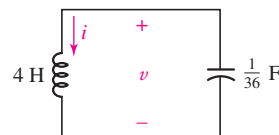
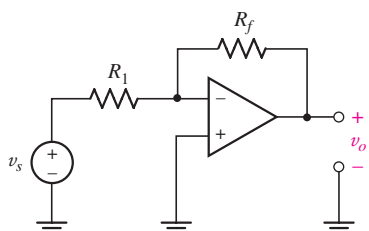


FIGURE 9.34 This circuit is lossless, and it provides the undamped response $v = 2 \sin 3t \text{ V}$, if $v(0) = 0$ and $i(0) = -\frac{1}{6} \text{ A}$.



■ FIGURE 9.35 The inverting operational amplifier provides a gain $v_o/v_s = -R_f/R_1$, assuming an ideal op amp.

In order to solve this equation, we plan to make use of the operational amplifier as an integrator. We assume that the highest-order derivative appearing in the differential equation here, d^2v/dt^2 , is available in our configuration of op amps at an arbitrary point A. We now make use of the integrator, with $RC = 1$, as discussed in Sec. 7.5. The input is d^2v/dt^2 , and the output must be $-dv/dt$, where the sign change results from using an inverting op amp configuration for the integrator. The initial value of dv/dt is 6 V/s, as we showed when we first analyzed the circuit, and thus an initial value of -6 V must be set in the integrator. The negative of the first derivative now forms the input to a second integrator. Its output is therefore $v(t)$, and the initial value is $v(0) = 0$. Now it only remains to multiply v by -9 to obtain the second derivative we assumed at point A. This is amplification by 9 with a sign change, and it is easily accomplished by using the op amp as an inverting amplifier.

Figure 9.35 shows the circuit of an inverting amplifier. For an ideal op amp, both the input current and the input voltage are zero. Thus, the current going “east” through R_1 is v_s/R_1 , while that traveling west through R_f is v_o/R_f . Since their sum is zero, we have

$$\frac{v_o}{v_s} = -\frac{R_f}{R_1}$$

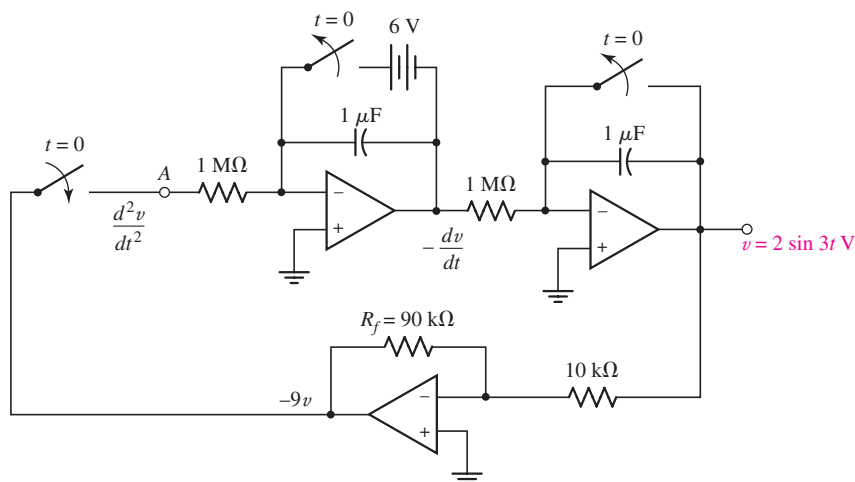
Thus, we can design for a gain of -9 by setting $R_f = 90$ k Ω and $R_1 = 10$ k Ω , for example.

If we let R be 1 M Ω and C be 1 μ F in each of the integrators, then

$$v_o = -\int_0^t v_s dt' + v_o(0)$$

in each case. The output of the inverting amplifier now forms the assumed input at point A, leading to the configuration of op amps shown in Fig. 9.36. If the left switch is closed at $t = 0$ while the two initial-condition switches are opened at the same time, the output of the second integrator will be the undamped sine wave $v = 2 \sin 3t$ V.

Note that both the LC circuit of Fig. 9.34 and the op amp circuit of Fig. 9.36 have the same output, but the op amp circuit does not contain a



■ FIGURE 9.36 Two integrators and an inverting amplifier are connected to provide the solution of the differential equation $d^2v/dt^2 = -9v$.

single inductor. It simply *acts* as though it contained an inductor, providing the appropriate sinusoidal voltage between its output terminal and ground. This can be a considerable practical or economic advantage in circuit design, as inductors are typically bulky, more costly than capacitors, and have more losses associated with them (and therefore are not as well approximated by the “ideal” model).



PRACTICE

9.11 Give new values for R_f and the two initial voltages in the circuit of Fig. 9.36 if the output represents the voltage $v(t)$ in the circuit of Fig. 9.37.

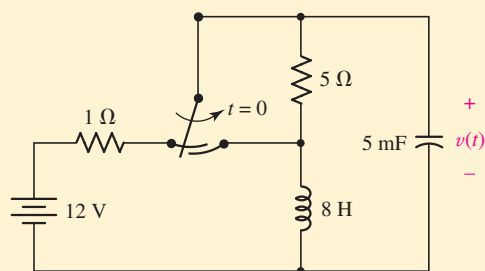


FIGURE 9.37

Ans: 250 kΩ; 400 V; 10 V.

SUMMARY AND REVIEW

The simple RL and RC circuits examined in Chap. 8 essentially did one of two things as the result of throwing a switch: *charge* or *discharge*. Which one happened was determined by the initial charge state of the energy storage element. In this chapter, we considered circuits that had two energy storage elements (a capacitor and an inductor), and found that things could get pretty interesting. There are two basic configurations of such RLC circuits: *parallel* connected and *series* connected. Analysis of such a circuit yields a *second-order* partial differential equation, consistent with the number of distinct energy storage elements (if we construct a circuit using only resistors and capacitors such that the capacitors cannot be combined using series/parallel techniques, we also obtain—eventually—a second-order partial differential equation).

Depending on the value of the resistance connected to our energy storage elements, we found the transient response of an RLC circuit could be either *overdamped* (decaying exponentially) or *underdamped* (decaying, but oscillatory), with a “special case” of *critically damped* which is difficult to achieve in practice. Oscillations can be useful (for example, in transmitting information over a wireless network) and not so useful (for example, in accidental feedback situations between an amplifier and a microphone at a concert). Although the oscillations are not sustained in the circuits we examined, we have at least seen one way to create them at will,

and design for a specific frequency of operation if so desired. We didn't end up spending a great deal of time with the series connected *RLC* circuit because with the exception of α , the equations are the same; only a minor adjustment in how we employ initial conditions to find the two unknown constants characterizing the transient response is needed. Along those lines, there were two "tricks," if you will, that we encountered. One is that to employ the second initial condition, we need to take the derivative of our response equation. The second is that whether we're employing KCL or KVL to make use of that initial condition, we're doing so at the instant that $t = 0$; appreciating this fact can simplify equations dramatically by setting $t = 0$ early.

We wrapped up the chapter by considering the *complete response*, and our approach to this did not differ significantly from what we did in Chap. 8. We closed with a brief section on a topic that might have occurred to us at some point—what happens when we remove the resistive losses completely (by setting parallel resistance to ∞ , or series resistance to 0)? We end up with an *LC* circuit, and we saw that we can approximate such an animal with an op amp circuit.

By now the reader is likely ready to finish reviewing key concepts of the chapter, so we'll stop here and list them, along with corresponding examples in the text.

- Circuits that contain two energy storage devices that cannot be combined using series-parallel combination techniques are described by a second-order differential equation.
- Series and parallel *RLC* circuits fall into one of three categories, depending on the relative values of R , L , and C :

Overdamped	$\alpha > \omega_0$
Critically damped	$\alpha = \omega_0$
Underdamped	$\alpha < \omega_0$

(Example 9.1)

- For series *RLC* circuits, $\alpha = R/2L$ and $\omega_0 = 1/\sqrt{LC}$. (Example 9.7)
- For parallel *RLC* circuits, $\alpha = 1/2RC$ and $\omega_0 = 1/\sqrt{LC}$. (Example 9.1)
- The typical form of an overdamped response is the sum of two exponential terms, one of which decays more quickly than the other: e.g., $A_1 e^{-t} + A_2 e^{-6t}$. (Examples 9.2, 9.3, 9.4)
- The typical form of a critically damped response is $e^{-\alpha t}(A_1 t + A_2)$. (Example 9.5)
- The typical form of an underdamped response is an exponentially damped sinusoid: $e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t)$. (Examples 9.6, 9.7, 9.8)
- During the transient response of an *RLC* circuit, energy is transferred between energy storage elements to the extent allowed by the resistive component of the circuit, which acts to dissipate the energy initially stored. (See Computer-Aided Analysis section.)
- The complete response is the sum of the forced and natural responses. In this case the total response must be determined before solving for the constants. (Examples 9.9, 9.10)

READING FURTHER

An excellent discussion of employing PSpice in the modeling of automotive suspension systems can be found in

R.W. Goody, *MicroSim PSpice for Windows*, vol. I, 2nd ed. Englewood Cliffs, N.J.: Prentice-Hall, 1998.

Many detailed descriptions of analogous networks can be found in Chap. 3 of

E. Weber, *Linear Transient Analysis Volume I*. New York: Wiley, 1954. (Out of print, but in many university libraries.)

EXERCISES

9.1 The Source-Free Parallel Circuit

- For a certain source-free parallel RLC circuit, $R = 1\text{ k}\Omega$, $C = 3\text{ }\mu\text{F}$, and L is such that the circuit response is overdamped. (a) Determine the value of L . (b) Write the equation for the voltage v across the resistor if it is known that $v(0^-) = 9\text{ V}$ and $dv/dt|_{t=0^+} = 2\text{ V/s}$.
- Element values of 10 mF and 2 nH are employed in the construction of a simple source-free parallel RLC circuit. (a) Select R so that the circuit is just barely overdamped. (b) Write the equation for the resistor current if its initial value is $i_R(0^+) = 13\text{ pA}$ and $di_E/dt|_{t=0^+} = 1\text{ nA/s}$.
- If a parallel RLC circuit is constructed from component values $C = 16\text{ mF}$ and $L = 1\text{ mH}$, choose R such that the circuit is (a) just barely overdamped; (b) just barely underdamped; (c) critically damped. (d) Does your answer for part (a) change if the resistor tolerance is 1%? 10%? (e) Increase the exponential damping coefficient for part (c) by 20%. Is the circuit now underdamped, overdamped, or still critically damped? *Explain*.
- Calculate α , ω_0 , s_1 , and s_2 for a source-free parallel RLC circuit if (a) $R = 4\text{ }\Omega$, $L = 2.22\text{ H}$, and $C = 12.5\text{ mF}$; (b) $L = 1\text{ nH}$, $C = 1\text{ pF}$, and R is 1% of the value required to make the circuit underdamped. (c) Calculate the damping ratio for the circuits of parts (a) and (b).
- You go to construct the circuit in Exercise 1, only to find no $1\text{ k}\Omega$ resistors. In fact, all you are able to locate in addition to the capacitor and inductor is a 1 meter long piece of 24 AWG soft solid copper wire. Connecting it in parallel to the two components you did find, compute the value of α , ω_0 , s_1 , and s_2 , and verify that the circuit is still overdamped.
- Consider a source-free parallel RLC circuit having $\alpha = 10^8\text{ s}^{-1}$, $\omega_0 = 10^3\text{ rad/s}$, and $\omega_0 L = 5\text{ }\Omega$. (a) Show that the stated units of $\omega_0 L$ are correct. (b) Compute s_1 and s_2 . (c) Write the general form of the natural response for the capacitor voltage. (d) By appropriate substitution, verify that your answer to part (c) is indeed a solution to Eq. [1] if the inductor and capacitor each initially store 1 mJ of energy, respectively.
- A parallel RLC circuit is constructed with $R = 500\text{ }\Omega$, $C = 10\text{ }\mu\text{F}$, and L such that it is critically damped. (a) Determine L . Is this value large or small for a printed-circuit board mounted component? (b) Add a resistor in parallel to the existing components such that the damping ratio is equal to 10. (c) Does increasing the damping ratio further lead to an overdamped, critically damped, or underdamped circuit? *Explain*.

9.2 The Overdamped Parallel RLC Circuit

- The circuit of Fig. 9.2 is modified substantially, with the resistor being replaced with a $1\text{ k}\Omega$ resistor, the inductor swapped out for a smaller 7 mH version, the capacitor replaced with a 1 nF alternative, and now the inductor is initially discharged while the capacitor is storing 7.2 mJ . (a) Compute α , ω_0 , s_1 , and s_2 , and verify that the circuit is still overdamped. (b) Obtain an expression

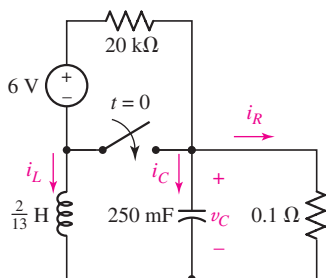


FIGURE 9.38

- for the current flowing through the resistor which is valid for $t > 0$. (c) Calculate the magnitude of the resistor current at $t = 10 \mu\text{s}$.
9. The voltage across a capacitor is found to be given by $v_C(t) = 10e^{-10t} - 5e^{-4t} \text{ V}$. (a) Sketch each of the two components over the range of $0 \leq t \leq 1.5 \text{ s}$. (b) Graph the capacitor voltage over the same time range.
10. The current flowing through a certain inductor is found to be given by $i_L(t) = 0.20e^{-2t} - 0.6e^{-3t} \text{ V}$. (a) Sketch each of the two components over the range of $0 \leq t \leq 1.5 \text{ s}$. (b) Graph the inductor current over the same time range. (c) Graph the energy remaining in the inductor over $0 \leq t \leq 1.5 \text{ s}$.
11. The current flowing through a 5Ω resistor in a source-free parallel RLC circuit is determined to be $i_R(t) = 2e^{-t} - 3e^{-8t} \text{ V}$, $t > 0$. Determine (a) the maximum current and the time at which it occurs; (b) the settling time; (c) the time t corresponding to the resistor absorbing 2.5 W of power.
12. For the circuit of Fig. 9.38, obtain an expression for $v_C(t)$ valid for all $t > 0$.
13. Consider the circuit depicted in Fig. 9.38. (a) Obtain an expression for $i_L(t)$ valid for all $t > 0$. (b) Obtain an expression for $i_R(t)$ valid for all $t > 0$. (c) Determine the settling time for both i_L and i_R .
14. With regard to the circuit represented in Fig. 9.39, determine (a) $i_C(0^-)$; (b) $i_L(0^-)$; (c) $i_R(0^-)$; (d) $v_C(0^-)$; (e) $i_C(0^+)$; (f) $i_L(0^+)$; (g) $i_R(0^+)$; (h) $v_C(0^+)$.

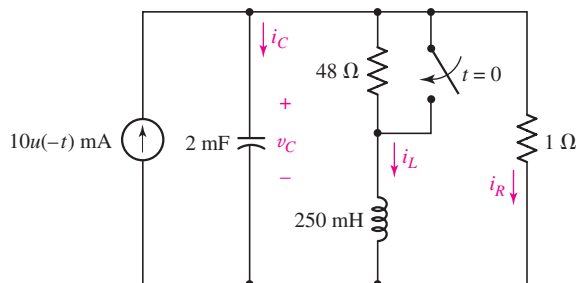


FIGURE 9.39

15. (a) Assuming the passive sign convention, obtain an expression for the voltage across the 1Ω resistor in the circuit of Fig. 9.39 which is valid for all $t > 0$. (b) Determine the settling time of the resistor voltage.
16. With regard to the circuit presented in Fig. 9.40, (a) obtain an expression for $v(t)$ which is valid for all $t > 0$; (b) calculate the maximum inductor current and identify the time at which it occurs; (c) determine the settling time.

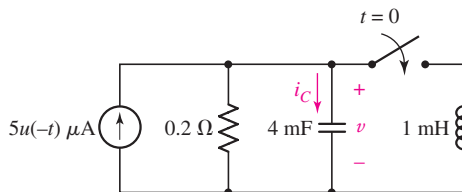


FIGURE 9.40

17. Obtain expressions for the current $i(t)$ and voltage $v(t)$ as labeled in the circuit of Fig. 9.41 which are valid for all $t > 0$.

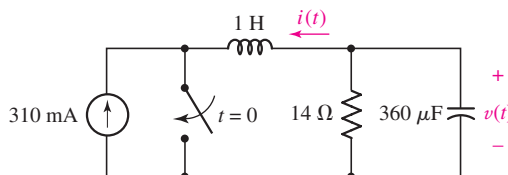


FIGURE 9.41



18. Replace the $14\ \Omega$ resistor in the circuit of Fig. 9.41 with a $1\ \Omega$ resistor. (a) Obtain an expression for the energy stored in the capacitor as a function of time, valid for $t > 0$. (b) Determine the time at which the energy in the capacitor has been reduced to one-half its maximum value. (c) Verify your answer with an appropriate PSpice simulation.



19. Design a complete source-free parallel RLC circuit which exhibits an overdamped response, has a settling time of 1 s, and has a damping ratio of 15.
20. For the circuit represented by Fig. 9.42, the two resistor values are $R_1 = 0.752\ \Omega$ and $R_2 = 1.268\ \Omega$, respectively. (a) Obtain an expression for the energy stored in the capacitor, valid for all $t > 0$; (b) determine the settling time of the current labeled i_A .

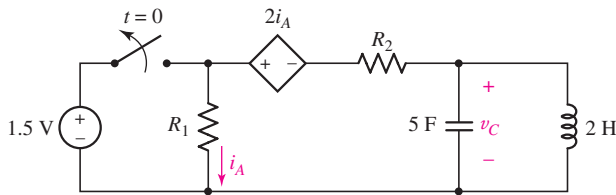


FIGURE 9.42

9.3 Critical Damping

21. A motor coil having an inductance of 8 H is in parallel with a $2\ \mu\text{F}$ capacitor and a resistor of unknown value. The response of the parallel combination is determined to be critically damped. (a) Determine the value of the resistor. (b) Compute α . (c) Write the equation for the current flowing into the resistor if the top node is labeled v , the bottom node is grounded, and $v = Ri_r$. (d) Verify that your equation is a solution to the circuit differential equation,

$$\frac{di_r}{dt} + 2\alpha \frac{di_r}{dt} + \alpha^2 i_r = 0$$

22. The condition for critical damping in an RLC circuit is that the resonant frequency ω_0 and the exponential damping factor α are equal. This leads to the relationship $L = 4R^2C$, which implies that $1\ \text{H} = 1\ \Omega^2 \cdot \text{F}$. Verify this equivalence by breaking down each of the three units to fundamental SI units (see Chap. 2).
23. A critically damped parallel RLC circuit is constructed from component values $40\ \Omega$, $8\ \text{nF}$, and $51.2\ \mu\text{H}$, respectively. (a) Verify that the circuit is indeed critically damped. (b) Explain why, in practice, the circuit once fabricated is unlikely to be truly critically damped. (c) The inductor initially stores 1 mJ of energy while the capacitor is initially discharged. Determine the magnitude of the capacitor voltage at $t = 500\ \text{ns}$, the maximum absolute capacitor voltage, and the settling time.



24. Design a complete (i.e., with all necessary switches or step function sources) parallel RLC circuit which has a critically damped response such that the capacitor voltage at $t = 1\ \text{s}$ is equal to 9 V and the circuit is source-free for all $t > 0$.
25. A critically damped parallel RLC circuit is constructed from component values $40\ \Omega$ and $2\ \text{pF}$. (a) Determine the value of L , taking care not to overround. (b) Explain why, in practice, the circuit once fabricated is unlikely to be truly critically damped. (c) The inductor initially stores no energy while the capacitor is initially storing 10 pJ. Determine the power absorbed by the resistor at $t = 2\ \text{ns}$, the maximum absolute inductor current $|i_L|$, and the settling time.

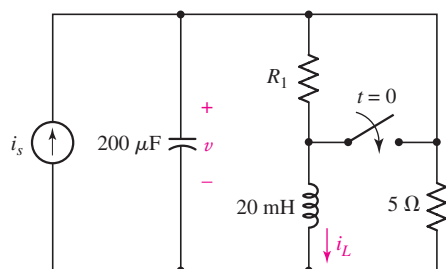


FIGURE 9.43

26. For the circuit of Fig. 9.43, $i_s(t) = 30u(-t)$ mA. (a) Select R_1 so that $v(0^+) = 6$ V. (b) Compute $v(2$ ms). (c) Determine the settling time of the capacitor voltage. (d) Is the inductor current settling time the same as your answer to part (c)?
27. The current source in Fig. 9.43 is $i_s(t) = 10u(1 - t)$ μ A. (a) Select R_1 such that $i_L(0^+) = 2$ μ A. Compute i_L at $t = 500$ ms and $t = 1.002$ ms.
28. The inductor in the circuit of Fig. 9.41 is changed such that the circuit response is now critically damped. (a) Determine the new inductor value. (b) Calculate the energy stored in both the inductor and the capacitor at $t = 10$ ms.
29. The circuit of Fig. 9.42 is rebuilt such that the quantity controlling the dependent source is now $100i_A$, a 2 μ F capacitor is used instead, and $R_1 = R_2 = 10$ Ω . (a) Calculate the inductor value required to obtain a critically damped response. (b) Determine the power being absorbed by R_2 at $t = 300$ μ s.

9.4 The Underdamped Parallel RLC Circuit

30. (a) With respect to the parallel RLC circuit, derive an expression for R in terms of C and L to ensure the response is underdamped. (b) If $C = 1$ nF and $L = 10$ mH, select R such that an underdamped response is (just barely) achieved. (c) If the damping ratio is increased, does the circuit become more or less underdamped? *Explain.* (d) Compute α and ω_d for the value of R you selected in part (b).
31. The circuit of Fig. 9.1 is constructed using component values 10 k Ω , 72 μ H, and 18 pF. (a) Compute α , ω_d , and ω_0 . Is the circuit overdamped, critically damped, or underdamped? (b) Write the form of the natural capacitor voltage response $v(t)$. (c) If the capacitor initially stores 1 nJ of energy, compute v at $t = 300$ ns.
32. The source-free circuit depicted in Fig. 9.1 is constructed using a 10 mH inductor, a 1 mF capacitor, and a 1.5 k Ω resistor. (a) Calculate α , ω_d , and ω_0 . (b) Write the equation which describes the current i for $t > 0$. (c) Determine the maximum value of i , and the time at which it occurs, if the inductor initially stores no energy and $v(0^-) = 9$ V.
33. (a) Graph the current i for the circuit described in Exercise 32 for resistor values 1.5 k Ω , 15 k Ω , and 150 k Ω . Make three separate graphs and be sure to extend the corresponding time axis to $6\pi/\omega_d$ in each case. (b) Determine the corresponding settling times.
34. Analyze the circuit described in Exercise 32 to find $v(t)$, $t > 0$, if R is equal to (a) 2 k Ω ; (b) 2 Ω . (c) Graph both responses over the range of $0 \leq t \leq 60$ ms. (d) Verify your answers with appropriate PSpice simulations.
35. For the circuit of Fig. 9.44, determine (a) $i_C(0^-)$; (b) $i_L(0^-)$; (c) $i_R(0^-)$; (d) $v_C(0^-)$; (e) $i_C(0^+)$; (f) $i_L(0^+)$; (g) $i_R(0^+)$; (h) $v_C(0^+)$.

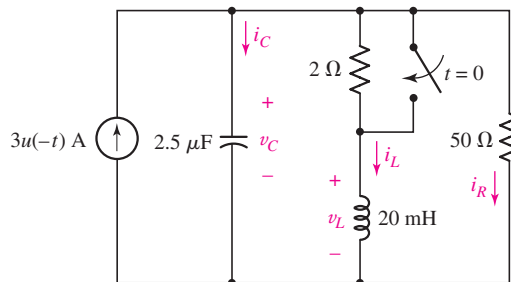


FIGURE 9.44

36. Obtain an expression for $v_L(t)$, $t > 0$, for the circuit shown in Fig. 9.44. Plot the waveform for at least two periods of oscillation.

37. For the circuit of Fig. 9.45, determine (a) the first time $t > 0$ when $v(t) = 0$; (b) the settling time.

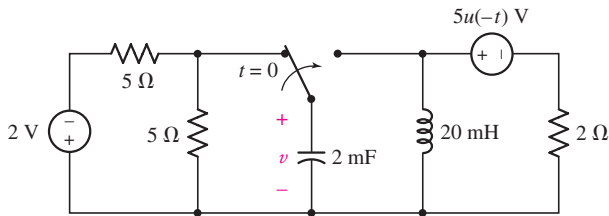


FIGURE 9.45

38. (a) Design a parallel RLC circuit that provides a capacitor voltage which oscillates with a frequency of 100 rad/s, with a maximum value of 10 V occurring at $t = 0$, and the second and third maxima both in excess of 6 V. (b) Verify your design with an appropriate PSpice simulation.
39. The circuit depicted in Fig. 9.46 is just barely underdamped. (a) Compute α and ω_d . (b) Obtain an expression for $i_L(t)$ valid for $t > 0$. (c) Determine how much energy is stored in the capacitor, and in the inductor, at $t = 200$ ms.

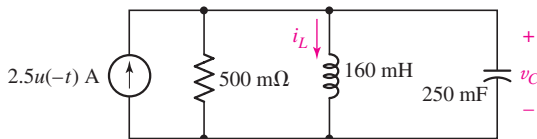


FIGURE 9.46

40. When constructing the circuit of Fig. 9.46, you inadvertently install a 500 MΩ resistor by mistake. (a) Compute α and ω_d . (b) Obtain an expression for $i_L(t)$ valid for $t > 0$. (c) Determine how long it takes for the energy stored in the inductor to reach 10% of its maximum value.

9.5 The Source-Free Series RLC Circuit

41. The circuit of Fig. 9.21a is constructed with a 160 mF capacitor and a 250 mH inductor. Determine the value of R needed to obtain (a) a critically damped response; (b) a “just barely” underdamped response. (c) Compare your answers to parts (a) and (b) if the circuit was a parallel RLC circuit.
42. Component values of $R = 2 \Omega$, $C = 1$ mF, and $L = 2$ mH are used to construct the circuit represented in Fig. 9.21a. If $v_C(0^-) = 1$ V and no current initially flows through the inductor, calculate $i(t)$ at $t = 1$ ms, 2 ms, and 3 ms.
43. The series RLC circuit described in Exercise 42 is modified slightly by adding a 2Ω resistor in parallel to the existing resistor. The initial capacitor voltage remains 1 V, and there is still no current flowing in the inductor prior to $t = 0$. (a) Calculate $v_C(t)$ at 4 ms. (b) Sketch $v_C(t)$ over the interval $0 \leq t \leq 10$ s.
44. The simple three-element series RLC circuit of Exercise 42 is constructed with the same component values, but the initial capacitor voltage $v_C(0^-) = 2$ V and the initial inductor current $i(0^-) = 1$ mA. (a) Obtain an expression for $i(t)$ valid for all $t > 0$. (b) Verify your solution with an appropriate simulation.
45. The series RLC circuit of Fig. 9.22 is constructed using $R = 1$ kΩ, $C = 2$ mF, and $L = 1$ mH. The initial capacitor voltage v_C is -4 V at $t = 0^-$. There is no current initially flowing through the inductor. (a) Obtain an expression for $v_C(t)$ valid for $t > 0$. (b) Sketch over $0 \leq t \leq 6 \mu$ s.

46. With reference to the circuit depicted in Fig. 9.47, calculate (a) α ; (b) ω_0 ; (c) $i(0^+)$; (d) $di/dt|_{0^+}$; (e) $i(t)$ at $t = 6$ s.

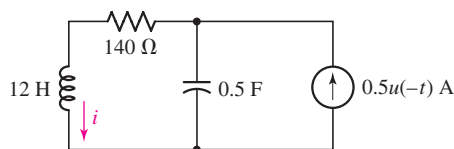


FIGURE 9.47

47. Obtain an equation for v_C as labeled in the circuit of Fig. 9.48 valid for all $t > 0$.

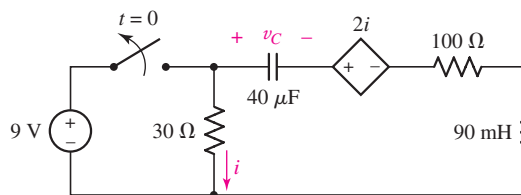


FIGURE 9.48



48. With reference to the series RLC circuit of Fig. 9.48, (a) obtain an expression for i , valid for $t > 0$; (b) calculate $i(0.8$ ms) and $i(4$ ms); (c) verify your answers to part (b) with an appropriate PSpice simulation.
49. Obtain an expression for i_1 as labeled in Fig. 9.49 which is valid for all $t > 0$.

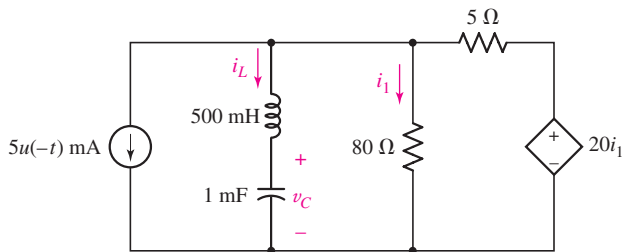


FIGURE 9.49

9.6 The Complete Response of the RLC Circuit

50. In the series circuit of Fig. 9.50, set $R = 1$ Ω. (a) Compute α and ω_0 . (b) If $i_s = 3u(-t) + 2u(t)$ mA, determine $v_R(0^-)$, $i_L(0^-)$, $v_C(0^-)$, $v_R(0^+)$, $i_L(0^+)$, $v_C(0^+)$, $i_L(\infty)$, and $v_C(\infty)$.

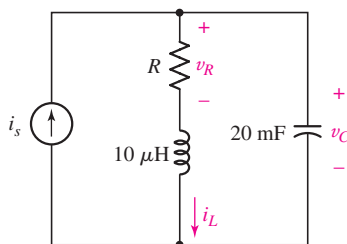


FIGURE 9.50

51. Evaluate the derivative of each current and voltage variable labeled in Fig. 9.51 at $t = 0^+$.

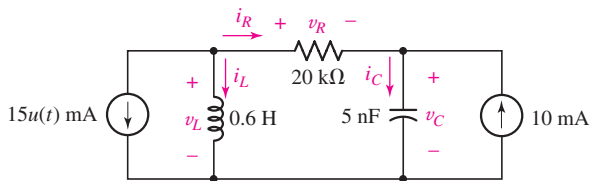


FIGURE 9.51

52. Consider the circuit depicted in Fig. 9.52. If $v_s(t) = -8 + 2u(t)$ V, determine (a) $v_C(0^+)$; (b) $i_L(0^+)$; (c) $v_C(\infty)$; (d) $v_C(t = 150 \text{ ms})$.
53. The 15Ω resistor in the circuit of Fig. 9.52 is replaced with a $500 \text{ m}\Omega$ alternative. If the source voltage is given by $v_s = 1 - 2u(t)$ V, determine (a) $i_L(0^+)$; (b) $v_C(0^+)$; (c) $i_L(\infty)$; (d) $v_C(4 \text{ ms})$.
54. In the circuit shown in Fig. 9.53, obtain an expression for i_L valid for all $t > 0$ if $i_1 = 8 - 10u(t)$ mA.
55. The 10Ω resistor in the series RLC circuit of Fig. 9.53 is replaced with a $1 \text{ k}\Omega$ resistor. The source $i_1 = 5u(t) - 4 \text{ mA}$. Obtain an expression for i_L valid for all $t > 0$.
56. For the circuit represented in Fig. 9.54, (a) obtain an expression for $v_C(t)$ valid for all $t > 0$. (b) Determine v_C at $t = 10 \text{ ms}$ and $t = 600 \text{ ms}$. (c) Verify your answers to part (b) with an appropriate PSpice simulation.

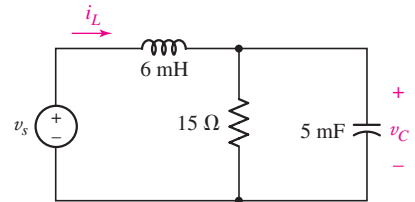


FIGURE 9.52

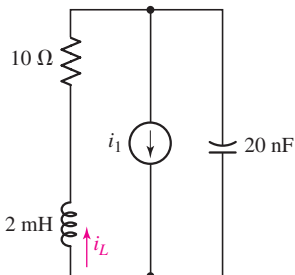


FIGURE 9.53

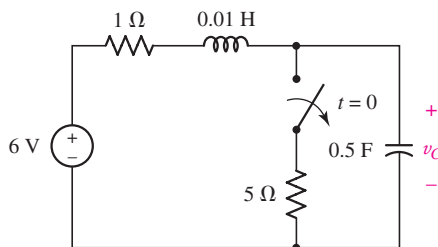


FIGURE 9.54

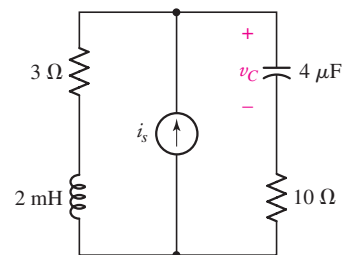


FIGURE 9.55

57. Replace the 1Ω resistor in Fig. 9.54 with a $100 \text{ m}\Omega$ resistor, and the 5Ω resistor with a $200 \text{ m}\Omega$ resistor. Assuming the passive sign convention, obtain an expression for the capacitor current which is valid for $t > 0$.
58. With regard to the circuit of Fig. 9.55, obtain an expression for v_C valid for $t \geq 0$ if $i_s(t) = 3u(-t) + 5u(t)$ mA.
59. (a) Adjust the value of the 3Ω resistor in the circuit represented in Fig. 9.55 to obtain a “just barely” overdamped response. (b) Determine the first instant ($t > 0$) at which an equal (and nonzero) amount of energy is stored in the capacitor and the inductor if $i_s(t) = 2u(t)$ A. (c) Calculate the corresponding energy. (d) At what subsequent time will the energy stored in the inductor be twice that stored in the capacitor at the same instant?

9.7 The Lossless LC Circuit

60. Design an op amp circuit to model the voltage response of the LC circuit shown in Fig. 9.56. Verify your design by simulating both the circuit of Fig. 9.56 and your circuit using an LF 411 op amp, assuming $v(0) = 0$ and $i(0) = 1 \text{ mA}$.

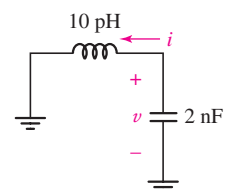


FIGURE 9.56

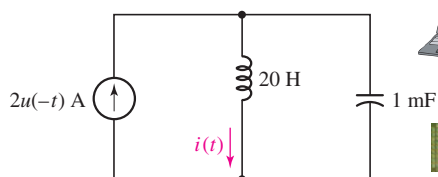


FIGURE 9.57



61. Refer to Fig. 9.57, and design an op amp circuit whose output will be $i(t)$ for $t > 0$.



62. Replace the capacitor in the circuit of Fig. 9.56 with a 20 H inductor in parallel with a $5 \mu\text{F}$ capacitor. Design an op amp circuit whose output will be $i(t)$ for $t > 0$. Verify your design by simulating both the capacitor-inductor circuit and your op amp circuit. Use an LM111 op amp in the PSpice simulation.



63. A source-free RC circuit is constructed using a $1 \text{ k}\Omega$ resistor and a 3.3 mF capacitor. The initial voltage across the capacitor is 1.2 V . (a) Write the differential equation for v , the voltage across the capacitor, for $t > 0$. (b) Design an op amp circuit that provides $v(t)$ as the output.

64. A source-free RL circuit contains a 20Ω resistor and a 5 H inductor. If the initial value of the inductor current is 2 A : (a) write the differential equation for i for $t > 0$, and (b) design an op amp integrator to provide $i(t)$ as the output, using $R_1 = 1 \text{ M}\Omega$ and $C_f = 1 \mu\text{F}$.

Chapter-Integrating Exercises

65. The capacitor in the circuit of Fig. 9.58 is set to 1 F . Determine $v_C(t)$ at (a) $t = -1 \text{ s}$; (b) $t = 0^+$; (c) $t = 20 \text{ s}$.

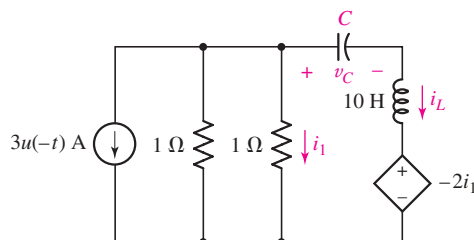


FIGURE 9.58

66. (a) What value of C for the circuit of Fig. 9.59 will result in an overdamped response? (b) Set $C = 1 \text{ F}$ and obtain an expression for $i_L(t)$ valid for $t > 0$.

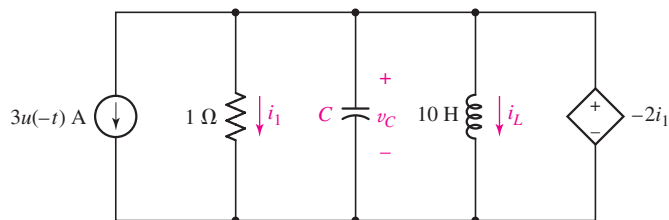


FIGURE 9.59

67. Obtain an expression for the current labeled i_1 in the circuit of Fig. 9.58 which is valid for $t > 0$, if the current source is replaced with a source $5u(t + 1) \text{ A}$.



68. Design a parallel RLC circuit which produces an exponentially damped sinusoidal pulse with a peak voltage of 1.5 V and at least two additional peaks with voltage magnitude greater than 0.8 V . Verify your design with an appropriate PSpice simulation.



69. Design a series RLC circuit which produces an exponentially damped sinusoidal pulse with a peak voltage of 1.5 V and at least two additional peaks with voltage magnitude greater than 0.8 V . Verify your design with an appropriate PSpice simulation.